ANxxx
PI7C9X3G606GP-808GP-816GP-1224GP-1632GP PCI Express® Gen3 Packet Switch Guidelines

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1 Introduction

This application note provides a concise and practical guide for designing a system board and carrier card with high speed signals that is easily incorporate the Pericom PCI Express Gen3 Packet Switches in their designs. This document includes design outlines, layout cautions and application reminders. It description stackup design, dielectric material selection and PCB fabrication technology used in the PCB design process.

This application note can help the designers to achieve optimal performance in their systems. This document is intended for audiences familiar with PCB manufacturing, layout, and design.

This document is not intended to be a specification. All information in the document is believed to be accurate as of publication date. However, no guarantees are given regarding the accuracy of this document.

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2 General Print Circuit Board Recommendations

Designing system board and carrier card that complies with the PCIe specification is ensuring the card thickness stays within 62 mil ± 5 mil. The thickness restriction significantly limits the total number of layers that can fit in.

2.1 Stackup Definition

The presented stack-ups in the following subsections are intended as examples which can be helping in stackup evaluation and selection. If a different stack-up is required other than those shown in the examples, please recalculate the dimensions of the traces. Work closely with your PCB manufacturer when selecting suitbale stack-up solution.

2.1.1 Four Layer Stack-up

The high speed signals on the layer 1 are referenced to the GND plane on layer 2. In this stack-up, it is preferential to route high speed signals on the layer 1 as opposed to the layer 4 so that the signals have a direct reference to the ground layer. For some designs it may be desirable to have the layer 4 as primary high speed routing layer. In this case, the power and ground usage on layer 2 and 3 coulde be swapped. The high speed signals should not be routed over two kind domain reference plane. Since the references for the high speed signals on the layer 4 are the power plane on layer 3, it is necessary to place stitching capacitors between the aforementioned power plane and ground.

Table 2-1 Four Layer Stack-up

PCIE 4-layer Stackup (Total : 62 mils, +/- 10%)			Layer	Type	Normally Material			High Speed Material(>=GEN3 recommand)		
Soldermask		1.2 mils			Material Name	Glass Style	εr (@3 GHz)	Material Name	Glass Style	εr (@10 GHz)
High Speed Signals		0.5 mils	1	Copper						
Prepreg		4.4 mils		Prepreg	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
GND Plane		1.4 mils	2	Copper						
Core	PTH Via	47 mils		Core	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
Power Plane		1.4 mils	3	Copper						
Prepreg		4.4 mils		Prepreg	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
High Speed Signals		0.5 mils	4	Copper						
Soldermask		1.2 mils								

2.1.2 Six Layer Stack-up

The high speed signals on the layer 1 are referenced to the power plane on layer 2. The signal reference for the layer 6 is the ground plane on layer 5. In this stack-up, it is preferable to route high speed signals on the layer 6. Power and ground layers could be swapped if it is desirable to have the primary high speed routing layer on the layer 1.

The reference planes for signals on layer 3 and 4 are located on layer 2 and 5. The referenced planes are on layers witch have a relatively large distance from layers 3 and 4. These layers are not suitable for routing high speed signals. In this stack-up approach, layers3 and 4 can only be useb for routing low speed signals where impedance matching is not required.

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Table 2-2 Six Layer Stack-up

PCIE 6-layer Stackup (Total : 62 mils, +/- 10%)			Layer	Type	Normally Material			High Speed Material(>=GEN3 recommand)		
Soldermask		1.0 mils			Material Name	Glass Style	εr (@3 GHz)	Material Name	Glass Style	εr (@10 GHz)
High Speed Signals	PTH Via	1.9 mils	1	Copper						
Prepreg		4.4 mils		Prepreg	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
Power Plane		1.4 mils	2	Copper						
ThinCore		4.3 mils		Core	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
Low Speed Signals		1.4 mils	3	Copper						
Prepreg		6.5 mils		Prepreg	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
Core		23 mils		Core	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
Prepreg		6.5 mils		Prepreg	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
Low Speed Signals		1.4 mils	4	Copper						
ThinCore		4.3 mils		Core	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
GND Plane		1.4 mils	5	Copper						
Prepreg		4.4 mils		Prepreg	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
High Speed Signals		1.9 mils	6	Copper						
Soldermask		1.0 mils								

2.1.3 Eight Layer Stack-up

The signals on the layer 1 and 8 are referenced to the plane in layer 2 and 7. The reference planes for signal layer 3 are the groud plane on layer 2 and the power plane on layer 4. When routing high speed signals on layer 3 and 6, it is necessary to place stitching capacitors between the aforementioned power plane and ground.

If layer 5 is rechange from power plane to groud plane. The inner layer 6 with the two adjacent ground planes is the best choice for routing high speed signals which have the most critical impedance control requirement. The inner layers caused less EMC problems as they are capsulated bye the adjacent ground planes. As layer 3 is referenced to a power plane, outer layer 1 and 8 are preferable for high speed routing if layer 6 is already occupied.

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Table 2-3 Eight Layer Stack-up

PCIE 8-layer Stackup (Total : 62 mils, +/- 10%)		Layer	Type	Normally Material			High Speed Material(>=GEN3 recommand)		
Soldermask	1.0 mils			Material Name	Glass Style	εr (@3 GHz)	Material Name	Glass Style	εr (@10 GHz)
High Speed Signals	2.0 mils	1	Copper						
Prepreg	3.7 mils		Prepreg	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
GND Plane	1.35 mils	2	Copper						
ThinCore	3.7 mils		Core	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
High Speed Signals	1.35 mils	3	Copper						
Prepreg	11.1 mils		Prepreg	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
Power Plane	2.70 mils	4	Copper						
Core	10 mils		Core	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
Power Plane	2.70mils	5	Copper						
Prepreg	11.1 mils		Prepreg	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
High Speed Signals	1.35 mils	6	Copper						
ThinCore	3.7 mils		Core	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
GND Plane	1.35 mils	7	Copper						
Prepreg	3.7 mils		Prepreg	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
High Speed Signals	2.0mils	8	Copper						
Soldermask	1.0 mils								

PTH Via

Note. Microstrip has a shorter propogation delay than stripline. It is recommended that TX and Rx be layered when routed, and change layers as few as possible.

2.1.4 Ten Layer Stack-up

The signals on the layer 1 and 10 are referenced to the plane in layer 2 and 9. The reference planes for signal layer 3 are the groud plane on layer 2 and the power plane on layer 4. The reference planes for signal layer 8 are the groud plane on layer 7 and the power plane on layer 9. When routing high speed signals on layer 3 and 8, it is necessary to place stitching capacitors between the aforementioned power plane and ground.

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Table 2-4 Ten Layer Stack-up

PCIE 10-layer Stackup (Total : 62 mils, +/- 10%)		Layer	Type	Normally Material			High Speed Material(>=GEN3 recommand)		
Soldermask	1.0 mils			Material Name	Glass Style	εr (@3 GHz)	Material Name	Glass Style	εr (@10 GHz)
High Speed Signals	1.95 mils	1	Copper						
Prepreg	3.65 mils		Prepreg	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
GND Plane	1.3 mils	2	Copper						
Core	6.2 mils		Core	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
High Speed Signals	1.3 mils	3	Copper						
Prepreg	6.2 mils		Prepreg	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
Power Plane	2.6 mils	4	Copper						
Core	4.5 mils		Core	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
High Speed Signals	1.3mils	5	Copper						
Prepreg	5.0 mils		Prepreg	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
High Speed Signals	1.3 mils	6	Copper						
Core	4.5 mils		Core	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
Power Plane	2.6 mils	7	Copper						
Prepreg	6.2 mils		Prepreg	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
High Speed Signals	1.3 mils	8	Copper						
Core	6.2 mils		Core	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
GND Plane	1.3 mils	9	Copper						
Prepreg	3.65 mils		Prepreg	FR4	1080	4.2	Isola FR408 Nelco N4000-12	2116 3313	3.67 3.7
High Speed Signals	1.95 mils	10	Copper						
Soldermask	1.0 mils								

PTH Via

2.2 Material Selection

A typical PCB stackup is comprised of alternate layers of copper foils separated by a dielectric material. The laminate/core is comprised of a thin dielectric material with copper clad foils bounded to both sides. These copper clad foils form the inner layers of the PCB. The core's dielectric material is comprised of cured (hardened) fiberglass-weave material with epoxy resin that acts as an insulation layer between the copper foils. The main purpose of the woven fiberglass weave is to provide mechanical strength for the PCB in both the X and Y directions. The woven glass fabric is available in different roll widths, sytles (1035, 1067, 1080, 1086, 2116, 3313, or 7628), and thicknesses.

Various resin systems are available for specific application requirements. Enhanced epoxy systems such as Getek, Megtron, 4000-13 and FR-408, which cite better performance than FR-4 with reduced Dk values and loss tangent (Df). High performance materials such as A-PPE by Asahi Glass, Nelco 600-21 Si, and Rogers 4350, which offer significant improvements in lower Dk, better impedance control, and reduced jitter.The parameters of interest are dielectric constant DK or ε_r, Loss tangent (Df) or dissipation factor, glass transition temperature requirements (Tg), ability to withstand soldering and rework, moisture absorption, and cost. The resin selected must be compatible with the glass fabric and the copper foil.

Unlike the laminate, a prepreg is a sheet of the dielectric material (fiberglass sheet saturated with resin) whose resin has not been fully cured. The prepreg acts as the insulation between core layers and is the gluing agent for the cores. The heat-press treatment cures the prepreg and binds all the layers together to form the PCB. The laminate vendors provide the laminate/core sheets in a wide variety of constructions such as different glass types for enhanced signal integrity performance, different fiberglass weaves, resin conten (%), core thickness, copper weights (1/2 oz, 1 oz, 2 oz), and copper foil types, among other considerations.

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The choice of electric material plays a significant role in determining the overall layer count. The key parameters to consider when selecting a dielectric material are:

- dielectric constant DK/ ε r
- loss tangent Df
- glass transition temperature (Tg)
- fiber weave characteristics
- Copper Foil Considerations

2.2.1 Dielectric Constant (Dk/ ε r)

Two of the most important factors that determine the electrical performance of a dielectric constant are impedance and signal integrity. The dielectric constant (Er) or relative permittivity (Dk) of a PCB material is generally between 3.5 and 5.5. A material's Er level depends on frequency and will usually drop as the frequency rises. The Dk level changes less on certain PCB materials than on others.

Theεr of a dielectric material that is specified in the material data sheet is the effectiveεr value based on a given fiberglass/resin composition because the glass and resin material have their ownεr values based on the type of glass and resin material chosen for creating the dielectric material. The effective DK orεr is used to calculate the impedance of the transmission lines.

Table 2-5 lists dielectric materials that are commonly used for PCB design. Most of the values listed below are specified at 10GHz or 12.5 GHz based on the manufacturer data sheet. Dielectric materials like Tachyon 100G or Megtron 7 are relatively new and are especially suitable for very high-speed desing althought at a higher cost.

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Table 2-5 Typical Dielectric Materials Used for PCB Design

Typical Dielectric Materials Used for PCB Design			
Material Name	Relative Dielectric Constant	Loss Tangent	Highest Data Sheet Frequency (GHz)
Gnenric FR4	4.5	0.015	N/A
Megtron 4	3.8	0.005	1
TU-872LKK	3.8	0.009	10
FR408HRIS	3.37	0.0092	10
EM-888	3.8	0.008	10
EM-888k	3.2	0.006	10
Nelco N4000-13	3.7	0.008	10
Nelco N4000-13 Si	3.3	0.007	10
Megtron 6	3.63	0.004	12
Rogers 4003C	3.38	0.0027	10
Rogers 4350B	3.48	0.0037	10
Taychon 100G	3.02	0.0021	10
Megtron 7 (Low DK glass)	3.35	0.002	12
I-Speed	3.63	0.0071	10
I-Speed IS	3.27	0.0064	10

The advantage of picking a dielectric material with a lower ε r compared to one with a higher ε r is that you can attain a higher trace impedance for a given trace geometry keeping everything else the same. You need to either plan for a significantly lower trace geometry or increase the dielectric thickness to achieve the same impedance when using a higher ε r dielectric material.

2.2.2 Loss Tangent (Df)

A material loses less power if its loss tangent is low. The dielectric loss tangent of materials used in printed circuit boards is generally in the range of 0.02 to 0.001. The former figure applies to the more widely used materials. However, the latter figure generally applies to high-end materials. tan (δ) also increases along with frequency. When it comes to digital circuitry, loss tangent is generally a factor of minor importance. Expectations to this rule would include any application where the frequency level exceeds 1Ghz. Loss tangent is more consequential for analog signals.

This parameter is a measure of the amount of RF energy absorbed by the laminate. The resin system has a separate loss tangent and so does the fiberglass cloth. The loss tangent specified in the material data sheet is a combination of the two. Similar to DK, the loss tangent tan (δ) or Df also varies with frequency and is dependent on the composition of the glass-to-resin ratio.

Many papers published in the industy quantify the impact of dielectric and conductor losses across a wide range of frequencies. It has been estimated that the ratio of conductor loss to the total loss was about 13% for an FR4-based substrate with Dk = 4.4 and Df = 0.02 to 20 GHz. However, the conductor loss as a percentage of the overall loss increased to around 30% when using a low-loss dielectric with a Dk = 3.7 and Df = 0.002. These findings clearly illustrate that the dielectric loss is the dominant loss mechanism to be concerned with when designing a PCB for high-speed applications.

Note: Picking a dielectric material with low Dk and low Df is recommended for the best signal integrity.

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2.2.3 Glass Transition Temperature (Tg)

A PCB substrate softens when exposed to temperatures above a certain threshold. The substrate hardens back to its natural state once removed from heat. The range at which this transformation can be applied to a substrate is known as glass transition temperature (Tg) — a unit noted in degrees Celsius. This is a critical parameter for PCB manufacturability because the z-axis expansion can cause plated through-holes to fracture due to excessive if the PCB soldering temperature exceeds the Tg of the dielectric material.

2.2.4 Fiber Weave Characteristics

The main purpose of the woven glass fabric as mentioned earlier is to provide mechanical strength in both X and Y directions for a PCB. The fiberglass weave manifests itself as variations in impedance and velocity along a trace if the distribution of the glass in the dielectric is not uniform. In a differential pair, this results in intra-pair skew between the P and N legs. This is more pronounced for data rates beyond 10Gb/ps because the amount of skew due to this effect can often be more than 10 to 15% of UI depending on the date rate and length of the channel. There has been lot of industry research in this area to quantify the impact of fiber weave on signal integrity performance and ways to mitigate its effect. A simple and effective solution to minigate this effect during the stackup definition process is to pick a laminate/prepreg material using a mechanically spread glass. The laminate manufacturer goes throuth the extra step of applying energy to enlarge and flatten the glass in both X/Y directions to make it more uniform. This application of energy has the effect of reducing variations in dielectric constant seen by the transmission lines at it traverses a PCB. Some ot the examples of this type of glass weave style are 1067MS, 1086MS, 1078MS, and 3313MS among others. The MS next to the glass style indicates mechanical spread.

Note: Pick laminate/preperg material using a mechanically spread glasses such as 3313MS to minimize the fiber weave effect.

2.2.5 Copper Foil Considerations

In addition to the dielectric material considerations that have been discussed, carefully consider the copper foil characteristics needed to achieve robust signal integrity performance for multi-gigabit channels.

There are two basic methods for manufacturing copper foils. These are rolled copper foils and electro-deposiited (ED) copper foils. The ED copper foil is the dominant method for manufacturing copper foils used in multilayer PCBs. The copper is typically designated by weight (1/2 oz., 1 oz., 2 oz.) and foil type.

- DSTF® (Drum Side Treated Foil): Adhesion treatment is applied to shiny/drum side
- RTF (Reverse Treated Foil): Same as DSTF®
- LP: Low Profile Foil with Tooth 5.1-9.9 Microns
- VLP: Very Low Profile Foil with Tooth < 5 Microns
- e-VLP/H-VLP: Very Low Profile Foils
- STD HTE (Standard Shiny Copper): Adhesion treatment is applied to matte side

There are specific advantages and disadvantages associated with each foil type. The standard ED/THE copper is not the most optimum foil to achieve robust impedance control and signal integrity performance for supporting multi-gigabit data rates, unlike the VLP foils, because of additional loss due to copper surface roughness. However, the downside with VLP foils is the relatively high cost and the low peel strength as opposed to other copper foils.

At high frequencies, the majority of the current distribution is pushed to the outer surface or “skin” of the trace due to skin effect. The amount of penntration into the metal, known as skin depth, is usually represented by the symbol δ. The skin depth is inversely related to the square root of frequency.
The skin depth is approximately 2 um (80 uin.) at 1 GHz. Hence, specifying 1/2 oz. copper (700 uin.) for signal traces as part of stackup design is more than sufficient for all high-speed links because the bulk of the current flows at the surface or bottom of the trace. The skin effect manifests itself primarily as an increase in resistance proportional to the square root of frequency. This is called conductor loss.

At very high frequencies (>1 GHz), the skin depth is on the same order as the typical surface roughness of the copper foils. This results in additional conductor loss because the copper routhness further impedes the current flow.

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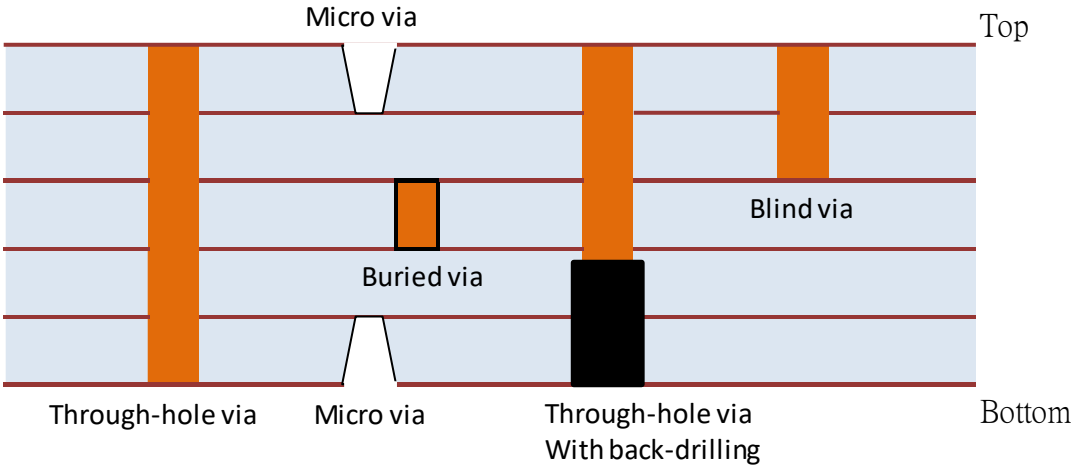
The roughness of the copper is dependent on the construction of the copper foils. The industry uses multiple terms (Ra, Rz, and RSAR) to specify copper roughness in um. Ra is defined as the average surface roughness, RSAR stands for roughness surface area ratio, and Rz is typically referred to as a ten-point height. The Rz value for a typical copper foil used in a PCB is around 6 to 10 um. This value is significantly higher than the skin depth at 1 GHz, making it imperative for you to account for conductor lossed due to copper surface roughness especially for PCB designs with transceiver channels operating at multi-gigabit data rates.

To address this issue, the laminate vendors provide laminate sheets with different copper foil constructions to cater to specific applications, although at a cost premium, RTF is the standard finish used by almost all laminate manufacturers. The Rz, Ra, and RSAR specifications for these foils are significantly different, resulting in differing amounts of conductor loss. It has been shown that you can expect at least a 1.5 dB improvement in conductor loss at 20 GHz moving from an RTF to a type of VLP foil. This change results in approximately a 17% improvement for overall conductor loss.

2.3 PCB Fabrication Technology

In addition to the material selection, the choice of PCB fabrication technology plays a key role in the number of signal layers needed for routing the system board and carrier card while meeting PCB thickness requirements. If your goal is to minimize the PCB layer count at the expence of cost, you can use advanced fabrication techniques like micro vias, blind vias, and buried vias in addition to thinner trace widths to achieve this goal. A brief description of various industry terms along with approximate cost adders on top of the standard PCB fabrication cost that can be expected for high-volume manufacturing is presented in the following section for reference. Figrue 1 shows the various via types.

Figure 1 Various via Types



2.3.1 Via Aspect Ratio

The Via aspect ratio is the ratio of PCB thickness to the smallest unplated via drill hole diameter. This is used as a guide to ensure that the PCB fabricator does not exceed the mechanical capabilities of the drilling equipment. A via aspect ratio of 10:1 is fairly common with standard PCB fabrication. The via aspect ratio can be increased to 20:1 using advanced PCB fabrication while maintaing design for manufacturing (DFM) rules.

2.3.2 Back-drilled Vias

A back-drilled via is a through-hole via that has a portion of its length drilled out so that it is no longer conductive. This improves signal integrity because it removes an unneeded stub from the route. The typical cost adder for back drilling vias ranges from 5% to 10 % of the total PCB fabrication cost.

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2.3.3 Via-in-Pad

A via-in-pad is a via drilled directly beneath a pad. This removes the need for a separate metal trace (stringer) to be drawn in order to drop down a via.This can help with breakout routing and improved signal integrity at the expence of a higher board fabrication cost. The cost adder varies from 10% to 15% of the PCB fabrication cost and is dependent on the via aspect ratio.

2.3.4 Blind and Buried Vias

A buried via is located entirely inside the PCB and does not touch the top or bottom layers. A blind via either travels from the top or bottom layer to an inner signal layer. This frees room above or below for other routing, unlike a through-hole via which travels all the way from the top to the bottom layer. The cost adder for a blind or a buried via depends on the number of different types of blind or buried vias that exist on the PCB. Each type of blind or buried via requies a separate lamination cycle, resulting in extra cost. For example, a PCB with three different type of blind or buried vias (L1 to L4, L6 to L2, L3 to L5) on a 10-layer PCB results in a +30% cost adder for each type of blind or buried via. The use of blind or buried vias is often called high-density interconnect (HDI).

The primary limitation of blind vias is the difficulty of plating copper in them that is thick enough and properly bonded to the inner layer to which they connect. The reliability of the plating becomes an issue when the depth of the blind via exceeds the diameter of the via (via aspect ratio limitation). Hence, most PCB fabrication shops recommend that the via diameter be at least 1.5 times the depth or greater to achieve a reliable plating. This limits blind vias to connecting only the top two layers of most PCBs. The blind vias must be stacked for connections below layer 2. This requires additional steps as part of the PCB fabrication process like “button” plating, which involes filling the blind vias with copper and using a sanding process to remove the excess copper that protrudes from some of these blind vias. This results in increased PCB fabrication cost.

2.3.5 Micro Vias

A micro via is a via whose diameter is less than 8 mil (0.2um). They are usually laser-drilled and typically cannot penetrate more than one or two layers at a time. The cost adder is approximately 15% for each type of via.

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3 SerDes Channel Guidelines

Some of the key recommendations for high-speed differential channels to ensure a robust layout from a signal and power integrity perspective include:

- Length-match the P and N legs for a given differential pair when doing the layout for PCIe interfaces.
- It is a good design practice to ensure that the stripline layers chosen for PCIe interfaces are surrounded by GND planes on either side to minimize coupling.
- Route RX and TX on separate stripline layers, giving higher priority to RX channels over TX channels from a layer assignment perpective.
- For a given AC coupling capacitor, it is recommended to use a smaller profile capacitor (0201 over a 0402 capacitor) to minimize the return loss on the RX channel. The smaller capacitor needs a smaller pad, thereby minimizing the pad capacitance resulting in a smaller impedance discontinuity with respect to the trace compared to a bigger capacitor. It is also recommended to route the via transitions to the AC coupling capacitors in true differential fashion on the P and N traces rather than in a single-ended fashion.
- It is a good design practice to remove all non-functional via pads for both the signal and GND vias to minimize the via capacitance. The via inductance can be reduced by minimizing the via stubs through back drilling. It is recommended to back drill the signal vias for all high-speed channels. The via impedance can be further optimized by adjusting the via antipad size through 3D simulation analysis along with ensuring a good return path by having the ground vias close to the signal vias. By optimizing the via impedance to closely match the trace impedance by reducing the via capacitance, inductance results in a significant improvement in the insertion and return loss performance across a wide range of frequencies.
- Breakout Areas near a device package that resulted in “neck-downs” and decreased spacing should be limited to no more than 500 mils in lengths. The necking-down should be done symmetrically on both nets of the differential pair. Breakout sections require special attention to minimize crosstalk.

- Test points and probing structures may impact the loss and jitter budgets. If possible, test points and probe structures should not introduce stubs on the differential pairs.

3.1 Trace Impedance

Care should be taken to distinguish between single-ended and differential trace impedance. High-speed differential pair signals such as PCIe, SATA, USB etc. need to be routed with differential impedance. This is impedance between the two signal traces of a pair. As the signals are referenced to ground, each differential pair signal also has single-ended impedance. When selecting trace geometry, priority should be given to matching the differential impedance over the single-ended impedance. The differential impedance is always smaller than twice the single-ended impedance.

$$Z_{Differential} < 2 \cdot Z_{Single\text{-}Ended}$$

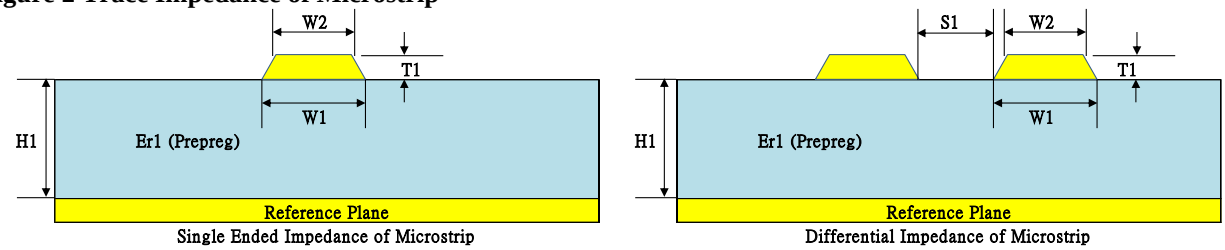
The signals allow a certain impedance tolerance (e.g. 50Ω +/- 15%). When defining trace geometry, try to keep the calculated impedance value as close as possible to the excat impedance value. This allows greater flexibility during PCB manufacture. Variation in impedances will occur between different production lots. If the calculated impedance is in the middle of the tolerance band, it will help ensure the maximum production yield.

Many PCB manufacturers use this tool. PCB manufacturers can often help customers with impedance calculations, and it is suggested that you work with chosen PCB manufactuter during you desing. Many PCB layout tools offer a very basic impedance calculator.

Microstrip is a somewhat simpler structure, with a single ground plane, a conductive trace, and a dielectric layer separating the signal conductor and ground plane. The following figure shows the geometry of such Microtrip. H1 is the distance from the trace to the according reference plane. Er1 is the relative permittivity of the isolation material. The traces have a trapezoid form due to the etching process. In the layout tool, the traces have to be designed with of W1. W2 depends on the trace height (T1) and the duration of the etching. Contact your PCB manufacturer in order to get the imformation about the resulting width W2. S1 is the spacing within a differential pair.

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Figure 2 Trace Impedance of Microstrip

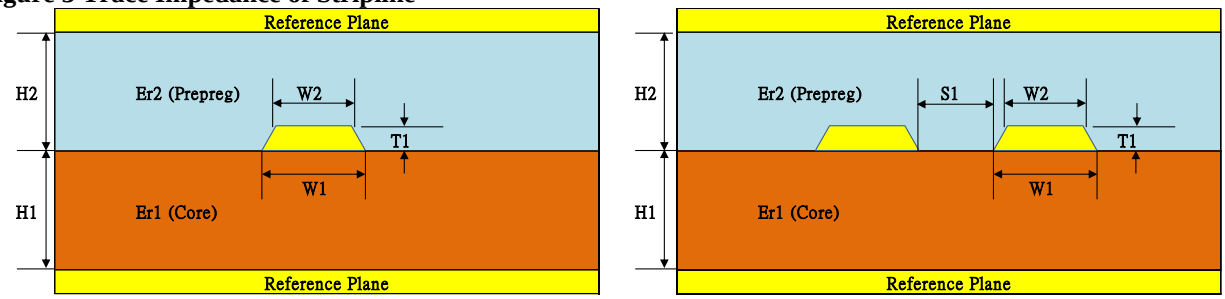


Single Ended Impedance of Microstrip

Differential Impedance of Microstrip

Stripline conductors are well isolated by the surrounding dielectric material. Trace in the inner layer of a PCB have two reference planes, reducing electromagnetic emissions and increasing immunity to external noise sources. The following figure shows the geometry of such striplines. When making impedance calculation of striplines, special care needs to be taken when it comes to the isolation thickness H1 and H2. H1 is the thickness of the core material. The traces are embedded in the prepreg material. As the traces have a finite height, the prepreg height H2 depends on the copper density. The relative permittivity of the core and prepreg material can be slightly different. Many impedance calculation tools can take this in account.

Figure 3 Trace Impedance of Stripline



Single Ended Impedance of stripline

Differential Impedance of stripline

The following table shows typical trace geometries for traces using the different layer stack-up

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Table 3-1 Typical Trace Geometries for Traces Using the Different Layer Stack-up

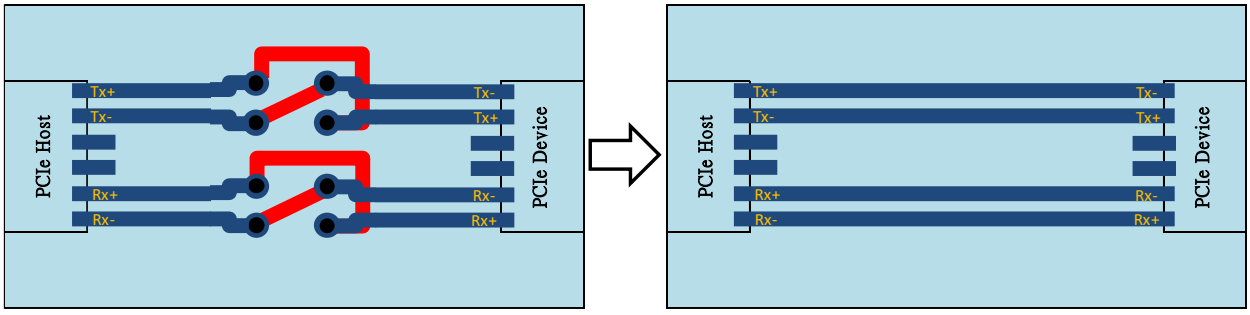
Typical trace geometries for traces using the four, six, eight layers stack-up						
Signal Type	Required $Z_{Single-Ended}$	Required $Z_{differential}$	Layer	Trace Type	W1	S1
Four layer stack-up						
Common Single Ended	50Ω	N/A	1,4	Microstrip	180um	N/A
PCIe, USB	50Ω	90Ω	1,4	Microstrip	180um	190um
SATA	55Ω	90Ω	1,4	Microstrip	150um	150um
Six layer stack-up						
Common Single Ended	50Ω	N/A	1,6	Microstrip	180um	N/A
PCIe, USB	50Ω	90Ω	1,6	Microstrip	180um	200um
SATA	55Ω	90Ω	1,6	Microstrip	150um	130um
Eight layer stack-up						
Common Single Ended	50Ω	N/A	1,8	Microstrip	180um	N/A
			3,6	Stripline	170um	N/A
PCIe, USB	50Ω	90Ω	1,8	Microstrip	180um	200um
			3,6	Stripline	170um	185um
SATA	55Ω	90Ω	1,8	Microstrip	150um	130um
			3,6	Stripline	140um	120um

3.2 Component Placement and Schematic Optimizations

The placement of the components is very often an underestimated subtask of the PCB design. Problems with signal return paths are very often related to suboptimal placement of the components. The high-speed signal connection should play a major role in the decisions that are taken during the placement. Certain signals only allow a certain amout of vias on the board. This means the number of layer changes should be kept as low as possible. If a signal needs to cross a splitting of its reference plane, one should first check whether this splitting is really unavoidable. Quite often, the solution can be a better placement of the components.

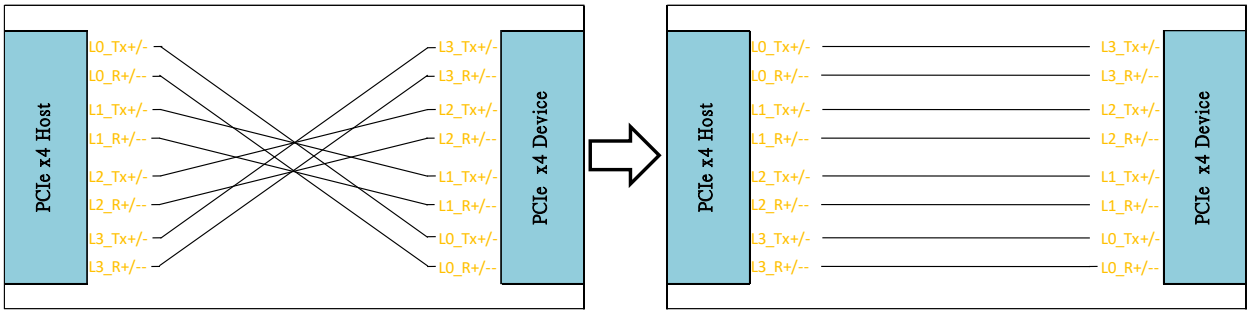
When placing the components, take in account that high-speed signals normally need to have more spacing than other signals. If there are different type of signals involed, like analogue signals and high-speed digital signals, placement of the components needs to be done very carefully.

When placing the components, try to avoid the need of crossing high-speed signals. Some interfaces, such as the PCIe, support polarity reversal. This means you can swap the positive and negative signal pins in order to avoid the need of complicatedly crossing the two traces of a differential pair signal.



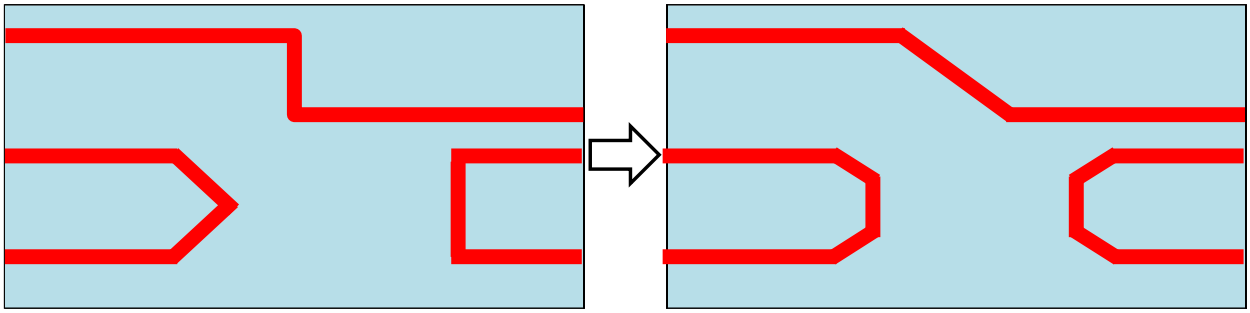
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Some of multi-lane interfacese support the lane reversal. This allows avoiding the crossing of the lanes. Please note that while polarity reversal is a mandatory feature of the PCIe interface, the lane reversal depends on the peripheral devices and host controller of the computer module. Please read carefully the appropriate datasheets before reversing the lanes.

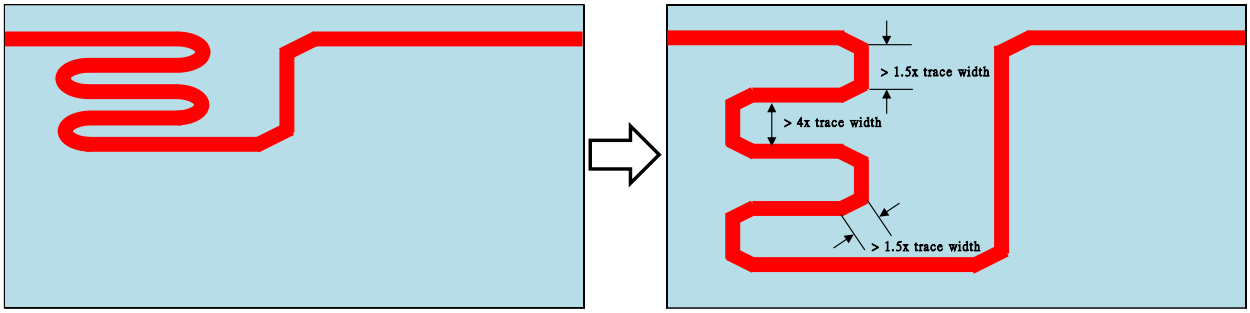


3.3 Trace Bend Geometry

When routing high-speed signals, bends should be minimized. If bends are needed, use 135° bends instead of 90° .



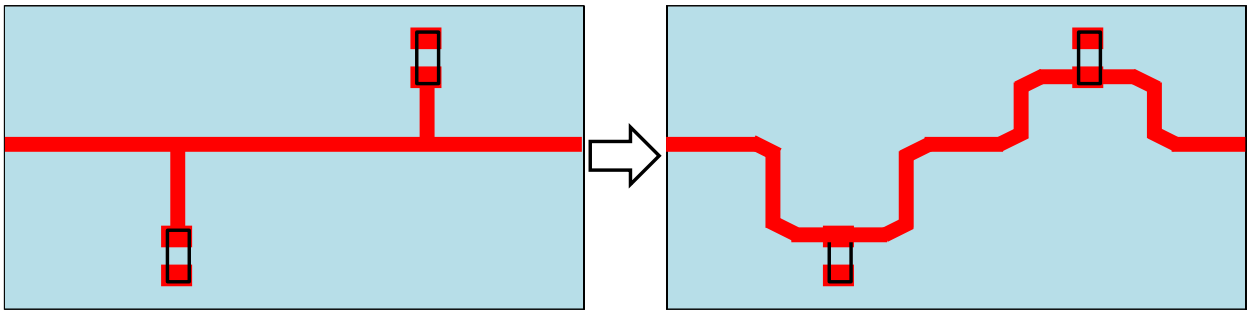
Serpentine traces (also called meander) are often needed when a certain trace length needs to be achieved. Keep a minimum distance of four times the trace width between adjacent copper in a single trace. The individual segments of the bends should be at least 1.5 times the trace width.



3.4 Trace Stubs

Long stub traces can act as antennas and stub traces can also produce reflections which negatively impacts signal interity. Common sources for stubs are pull-up or pull-down resistors on high speed signals. If such resistors are required, route the siganls as a daisy chain.

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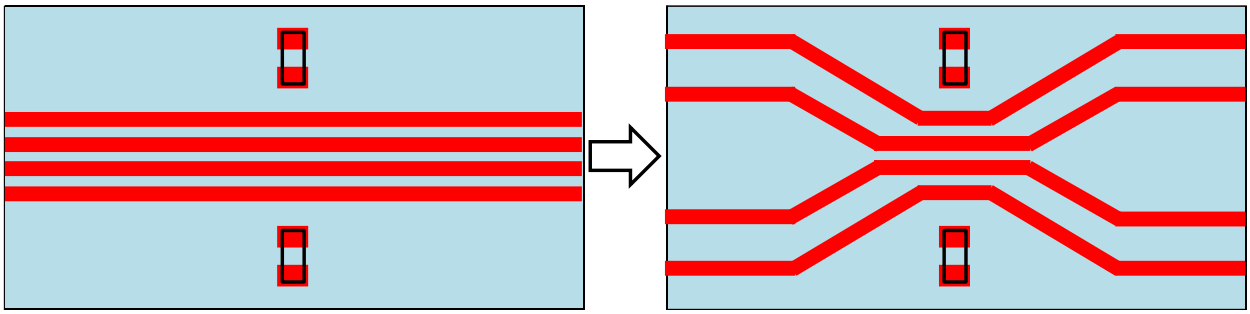


Vias can also act as stubs. For example, when a signal changes form layer 1 to 3 by using a via, the via creates a stub. As a rule of thumb, stubs longer than a tenth of the wavelength should be considered as problematic. Back-drilling the vias in order to avoid such stubs is a quite expensive technology and one which is not supported by most PCB manufactures. The only practical solution is to reduce the number of vias in high-speed traces. The following example shows the calculations on a Gen3 PCIe signal:

$$l_{MAX} \ll \frac{1}{10}\lambda_{MIN} = \frac{v}{10 \bullet f_{MAX}} = \frac{\frac{c}{\sqrt{\varepsilon r}}}{10 \bullet f_{MAX}} = \frac{\frac{300 \bullet 10^6\, m/s}{\sqrt{4.5}}}{10 \bullet 4GHz} = 3.5\text{mm}$$

3.5 Signal Proximity

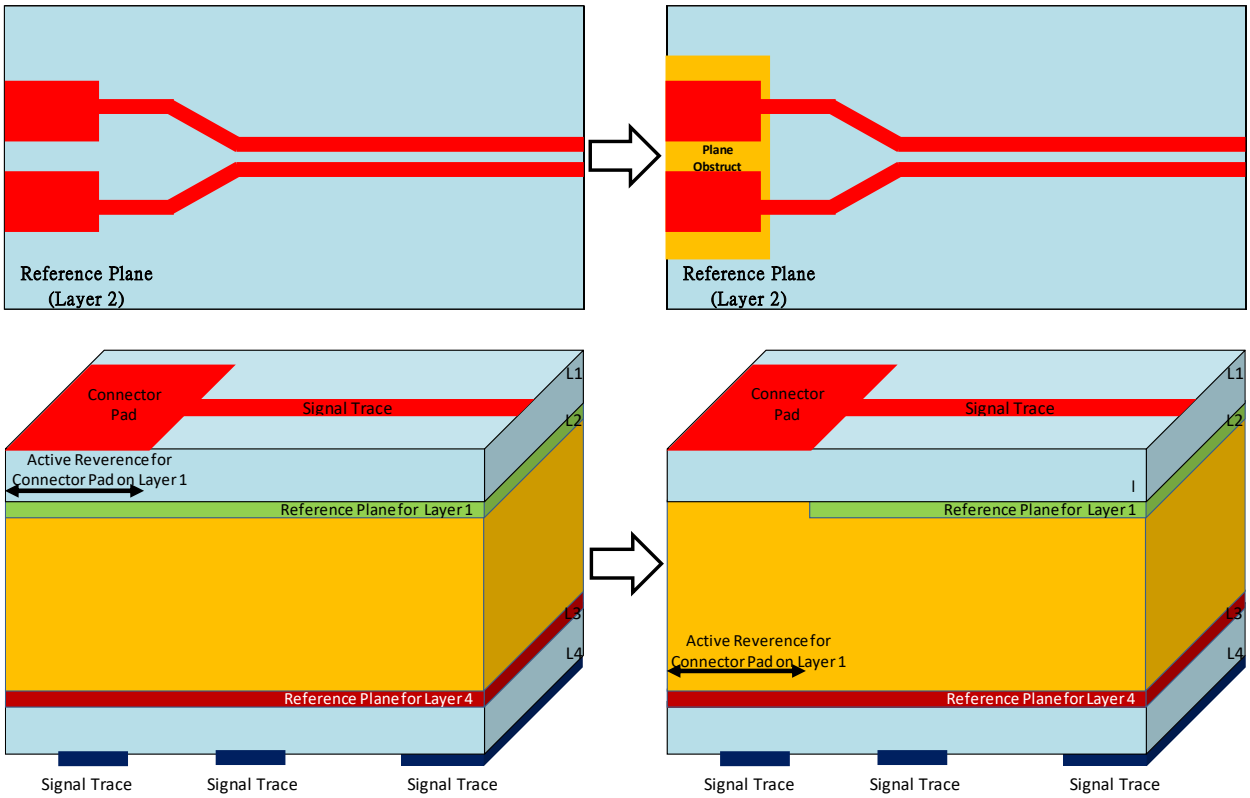
A minimum distance is required in order to minimize crosstalk between traces. The level of crosstalk depends on the distance between two traces and the length in which there are closely routed. Sometimes, bottlenecks can force the routing of traces closer than to what is normally permitted. Try to minimize such areas and enlarge the distance between the signals outside the bottleneck. If thear is space available, try to enlarge the distance between the high-speed signals (and between high-speed and low-speed signals) even if the minimum trace separation requirement has been met.



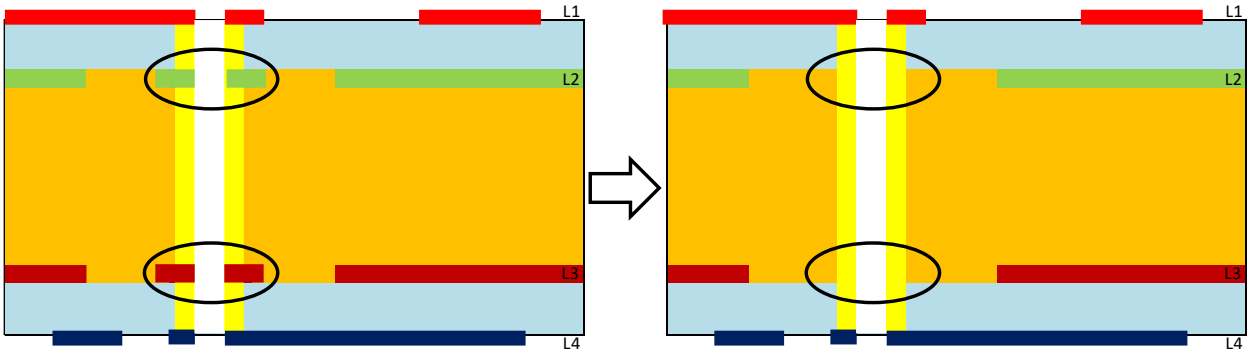
3.6 Ground Planes under Pads

The impedance of a trace depends on its width and the distance between trace and reference plane. A wide trace has lower impedance than a thin one with the same diftance. The same effect also exists for connector and component pads. A large pad has significantly lower impedance than the trace which is connected to the pad. This impedance discontinuity can cause reflections reduces signal integrity. In addition, A large pad has Capacity Effect.
Therefore, the reference planes on the inner layers should be removed immediately under the chip pin, AC capacitor and slot solder spots along the differential pair signal traces.

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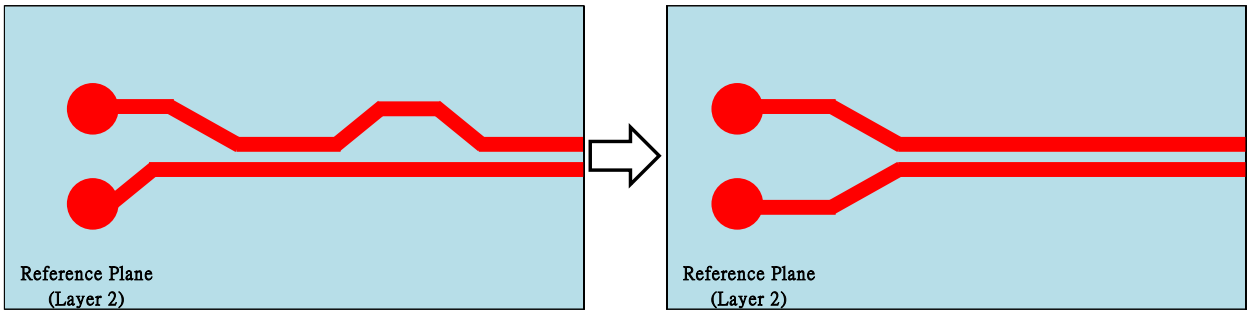
Vias are another source of impedance discontinuity. In order to minimize the effect, the unused pads of vias in inner layers should be removed. This can be done at design time in the CAD tool or by the PCB manufacturer.



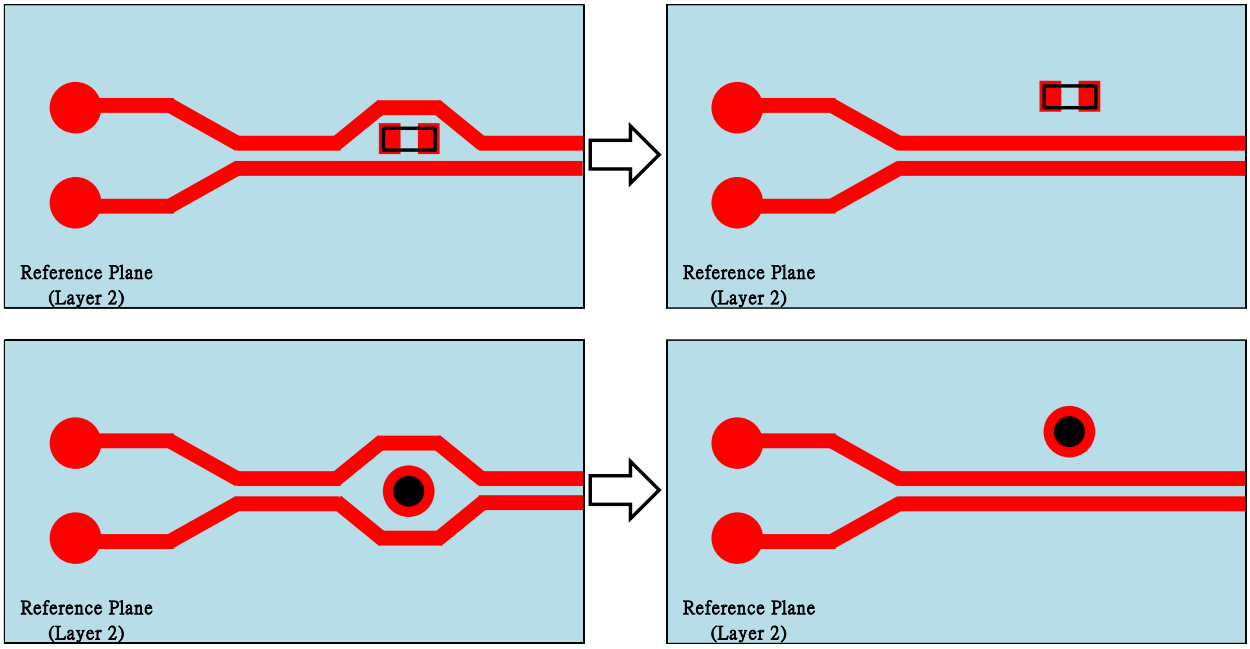
3.7 Differential Pair Signals

Differential pari signals need to be routed symmetrically. Try to minimize the area in which the specified spacing is enlarged due to pad entries.

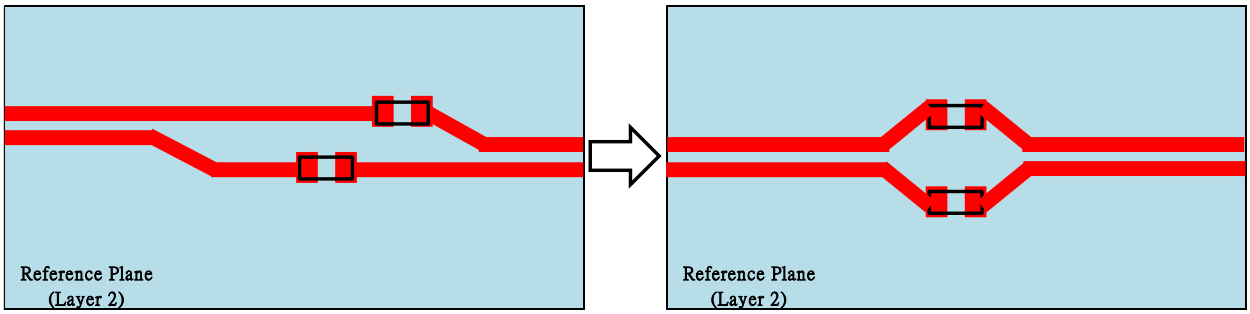
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Components and vias between the pairs could lead to EMC compliance problems and create an impedance discontinuity. So, it is not permitted to place any components or vias between the differential pairs, even if the signals are routed symmetrically.

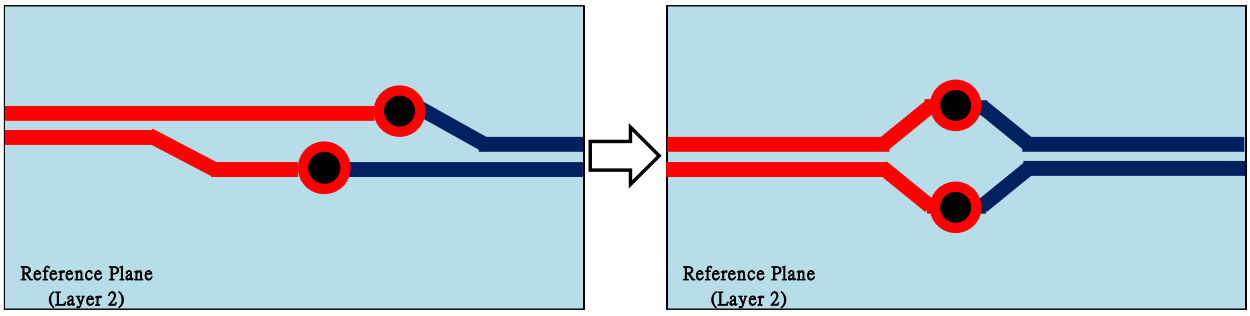


Some differential pair high-speed signals require serial coupling capacitors. Place such capacitors symmetrically. The capacitors and the pads create impedance discontinuities. 0201 sized capacitors are preferable, 0402 are acceptable. Do not palce larger packages such as 0603, 0805 or C-packs.

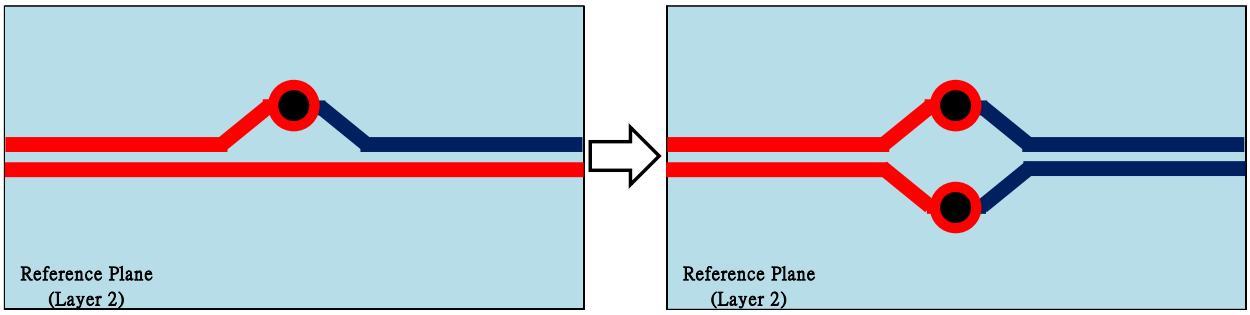


Vias introduce a huge discontinuity in impedance. Try to reduce the amount of placed vias to a minimum and palce the vias symmetrically.

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In order to meet the impedance requirements of a differential pair, both signal traces need to be routed on the same layer. Add the same amount of vias to the traces.



3.8 Length Matching

High-speed interfaces have additional requirements regarding the time of arrival skew between differnet traces and pairs of signals. All data siganls need to arrive within a time period in order to meet the setup and hold time requirements of the receiver. The board designer needs to make sure that such permitted skew is not exceeded. In order to meet this requirement, length matching is required. Often, the requirements are given as a maximum time skew. In order to calculate the maximum trace length difference, the propagation speed on the PCB needs to be estimated. The following formula can be used for calculating the speed:

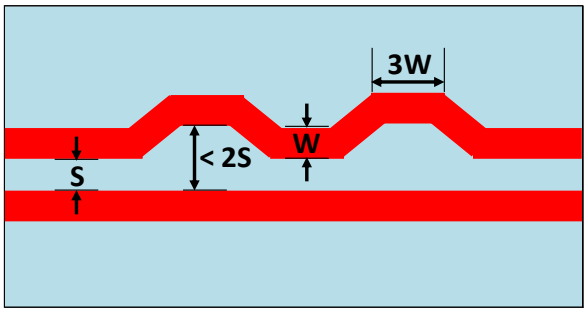
$$v = \frac{c}{\sqrt{\varepsilon r}}$$

The symbol c stands for the speed of light while ε r is the relative permittivity of material between the trace and the reference plane. The relative permittivity of an FR-4 PCB is around 4.5 while air has 1. A few of the coupling between Microstrip signals on the outer layer of a PCB and the reference plane is over the air and the solder mask. Since the relative permittivity of both materials is lower than the one of the FR-4, the signals are propagated slightly faster than in the inner layers (Striplines).εr As a rule of thumb, the signals on a PCB are propagated at half the speed of light. This equals to a speed of around 150 um/ps.

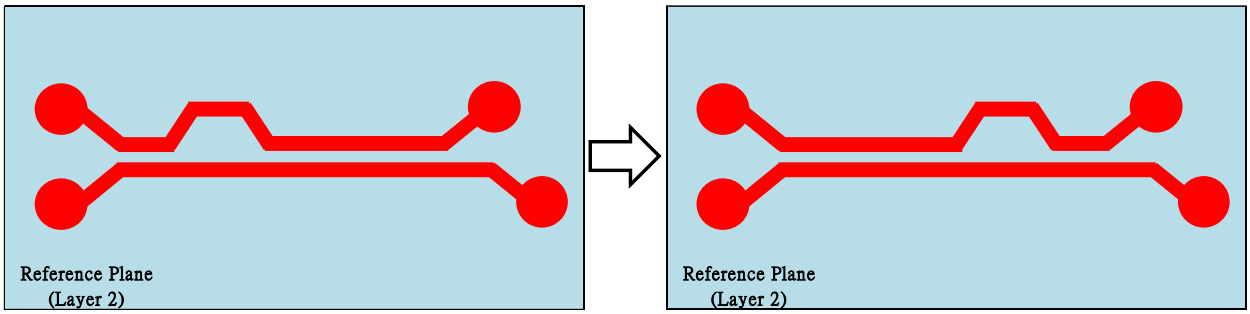
$$v \approx 150 \frac{um}{ps}$$

Differential pair signals often require a very tight delay skew between the positive and negative signal traces. Therefore, length differences need to be compensated for using serpentines (also called meanders). The geometry of serpentine traces need to be carefully chosen in order to reduce impedance discontinuity. The following figure shows the requirements for ideal serpentine traces:

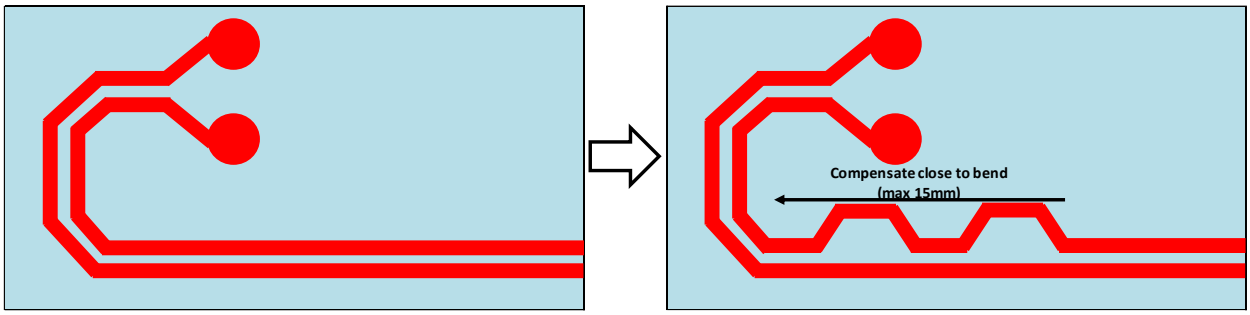
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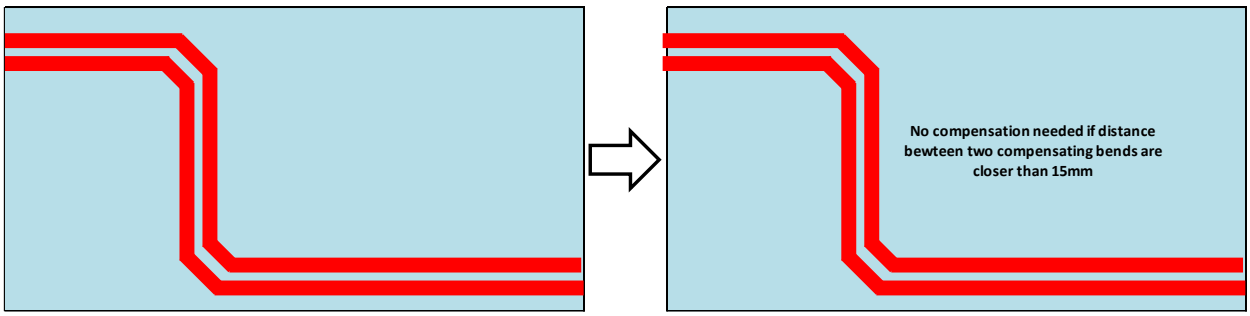
The serpentine traces should be placed at the origin of the length mismatching. This ensures that the positive and negative signal components are propagated synchronously over the major part of the connection.



Bends are a common source of length mismatching. The compensation should be placed close to the bend with maximum distance of 15mm.



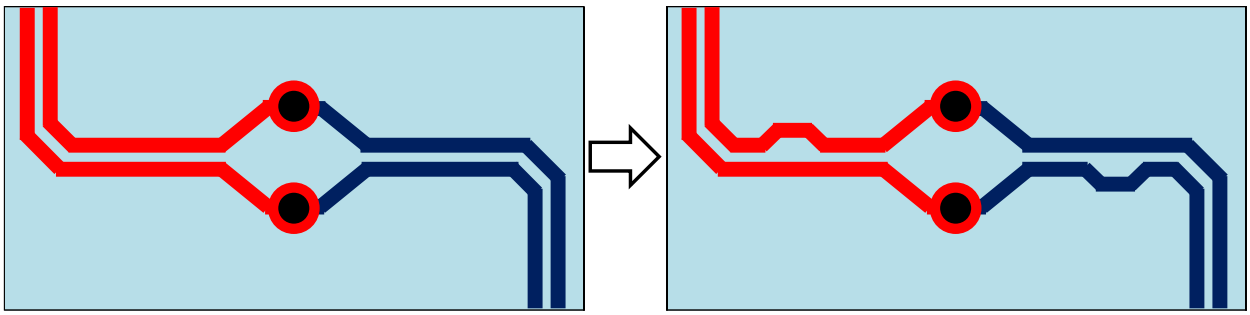
Often two bends compensate each other. If the bends are closer than 15mm, no additional compensation with serpentines is necessary. The signals should not propagate asynchronously over a distance greater than 15mm.



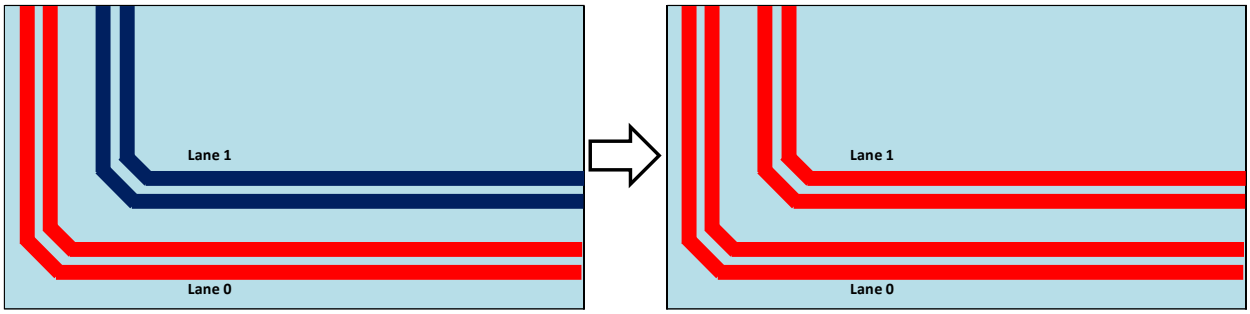
Each segment of a differential pair connection needs to be matched individually. A connection can be segmented by a connector, serial coupling capacitors or vias. The two bends in the following figure would compensate each other. Since

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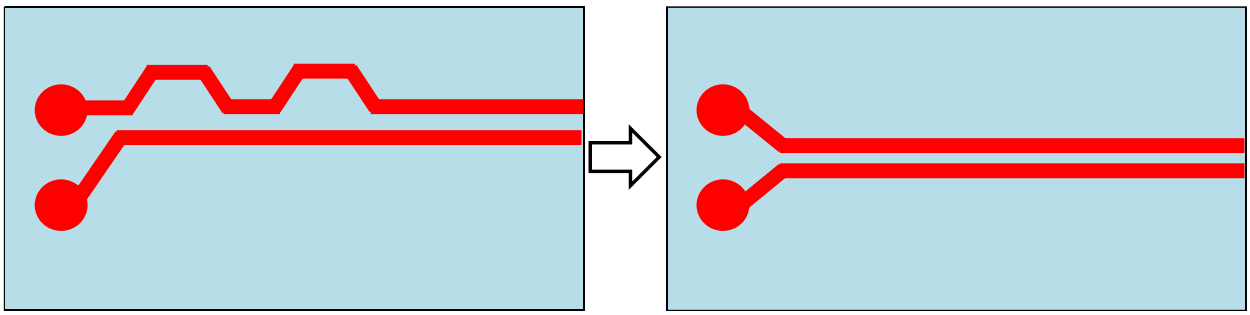
the vias divide the diffenential pair into two segments, the bends need to be compensated individually. This makes sure that the positive and negative signals are propagated synchronously through the vias.



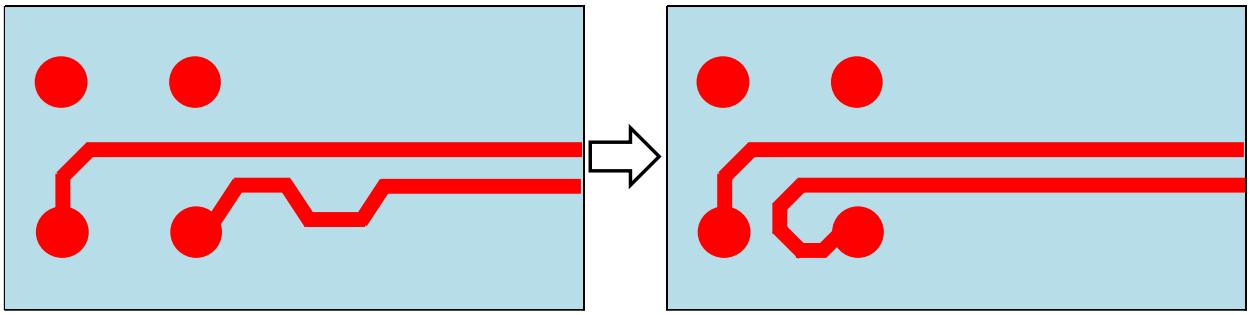
The PCIe interface requires tight matching between the lane to lane signal pairs. The signal speed is not equal for different layers. Since the difference is hard to estimate, it is preferable to route signals on the same layer if they need to be match.



Whenever possible, a symmetric breakout of differential pair signal is preferred in order to avoid the need of serpentine traces.



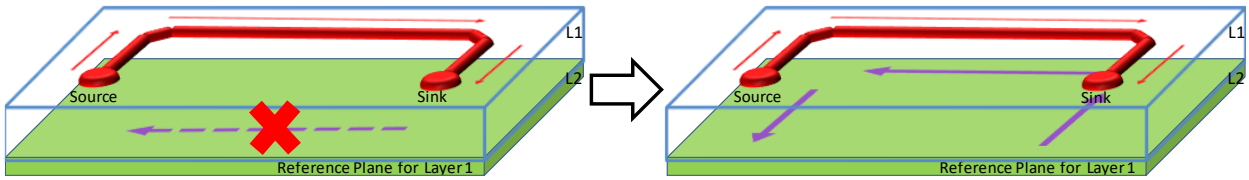
If the same between the pads permits, try to add a small loop to the shorter trace. As opposed to creating a serpentine trace.



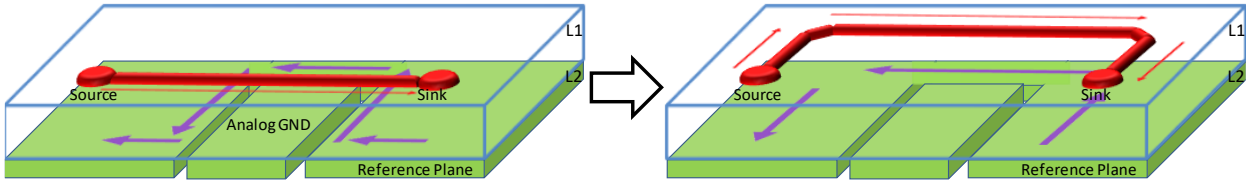
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3.9 Signal Return Path

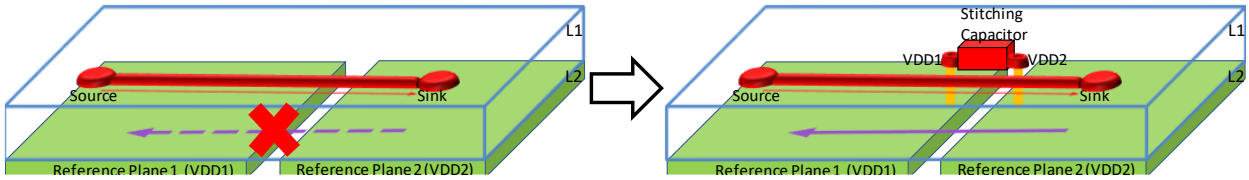
The return path of the signal current should always be considered when routing a signal. The return current of high speed signals tries to follow the signal path. Differential pair signals feature a positive and negative signal trace. Even these signals require a return path which needs to be considered when routing such signals. An incorrect signal return path is one of the most common sources for noise coupling and EMI problems.



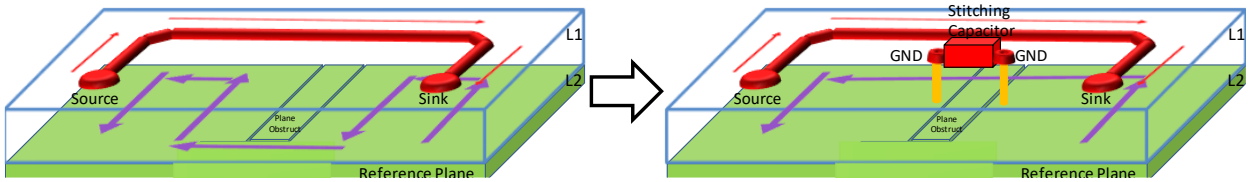
A signal should not be routed over a spit plane as the return path is not able to follow the signal trace. If a plane is split between a sink and source, route the signal trace around it. If the forward and return paths of a signal are separated, the area between them acts as a loop antenna.



If a signal needs to be routed over two diffenent reference planes, a stitching capacitor between the two reference plance is needed. The capacitor allows the return current to travel from one reference plane to the other. The capacitor should be placed close to the signal path in order to keep the distance between forward and return path small. A good value for the stitching capacitor is between 10nF and 100nF.

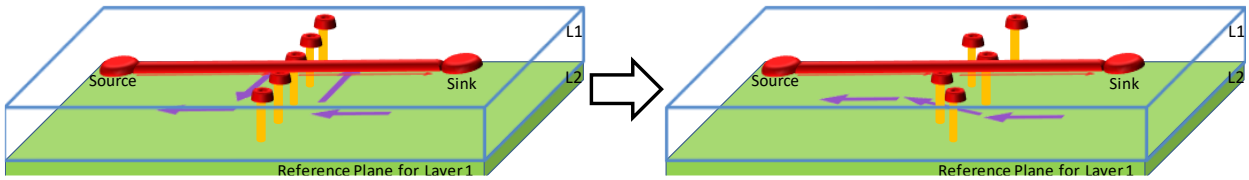


Plane obstrctions and plane slots should be avoided in general. Try to avoid routing signals over such obstrcutions. If unavoidable, stitching capacitors should be used to minimize the problems created by a separated return path.

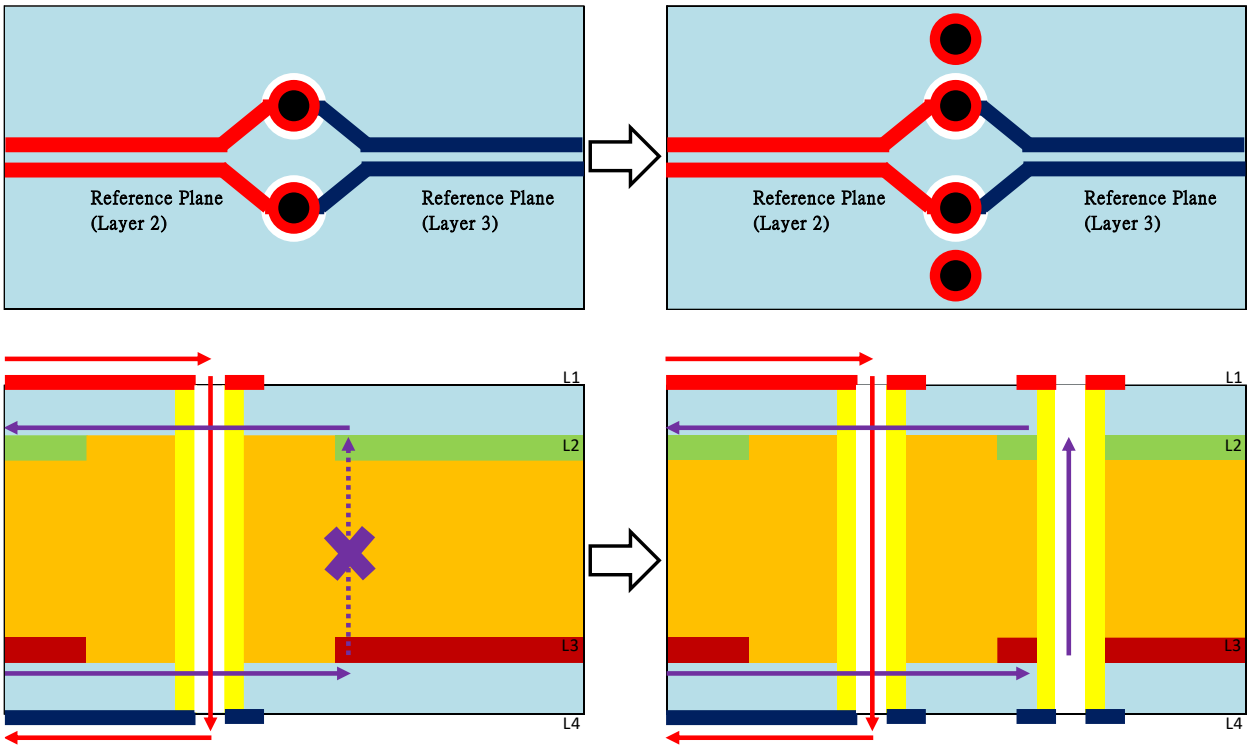


Voids in reference planes can result when placing vias close together. Be aware of such voids when routing high-speed signals. Try to avoid large void areas by ensuring adequate separation between vias. Sometimes, it is better to place fewer ground and power vias in order to reduce via voids.

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If a signal trace switches layer, stitching vias should to be added close to the layer change vias. This allows the return current to change ground plane. For differential signals, stitching groud vias should be placed symmetrically.



3.10 Total loss budgets

Some guides on routing will define a trace length maximum, either as a hard number or as a range. Total loss budgets in the table below; these values are taken at the maximum data rate specified for each generation. Note that these budgets are inclusive of insertion, return, connector, and dielectric/roughness losses along the length of a PCIe lane.Detailed information can be found in PCI_Express_CEM_r3.0

Table 3-2 Total loss budgets

Generation	Loss Budget	Differential Impedance.
Gen1	13.2 dB	100 Ohms
Gen2	16.5 dB	85 Ohms
Gen3	20 dB	85 Ohms

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4 Reduction of Crosstalk in High Speed Material PCB

In high frequency circuits, crosstalk is probably the most difficult factor to understand and predict.
One of the worries in layout design is the interference of edge coupling to the line characteristics, among which the effect caused by FEXT is the most terrible. FEXT has a shorter time and stronger energy characteristics, which is easy to interfere with a large number of signals at a time, making the receiving chip eye diagram worse, and more likely to generate misjudgment.
However, crosstalk can be controlled or even eliminated. The following strategies can be adopted.

4.1 Line Spacing

To reduce the coupling energy, the simplest and most direct way is to reduce the mutual capacitance and mutual inductance between the lines, and to achieve this goal, simply open the distance between the two lines. In almost all cases, separating the coupled lines can greatly reduce crosstalk interference. It's not easy to increase the distance between coupled lines. If your wiring is very dense, you have to spend a lot of effort to reduce the wiring density. If you are concerned about crosstalk interference, you can add one or two isolation layers. Line width and thickness also affect crosstalk, but their effect is much smaller than the distance factor. Therefore, these two parameters are generally rarely adjusted.

4.2 Adding Guard Trace

In fact, Guard trace is a type of grounding method. The concept is to add a ground path between the wire and the wire, because this path is much longer compared to the ground plane, very similar to the wire, and can block the electromagnetic field to interfere with the adjacent wire, so term term Guard trace is given, but the premise is that the design of Guard trace should be perfect.

4.3 Reducing the Thickness of Soldermask

When lines are routed on the PCB in the form of microstrip lines, the coupling path is not only from the lower dielectric that interferes with the adjacent lines, but also from the upper dielectric. This dielectric, called Soldermask, is used to prevent oxidation of the metal in the alignment. When we reduce its thickness, the coupling of the electric field can be reduced, and the coupling noise will naturally be reduced. However, reducing the thickness of the Soldermask makes a significant improvement in NEXT, but the effect on FEXT is rather insignificant, and it can even be said that there is hardly any difference in FEXT.

4.4 Layout in the Inner Layer

Compared to microstrip lines, the theoretical value of the coupling of striplines in FEXT is 0, but this is of course the result of an ideal ribbon line. In reality, the thickness of the board and the DK value of the material are unlikely to be the same for the upper and lower layers, so it is impossible to have no FEXT at all, and lines that are more sensitive to noise can be laid out in the inner layer to significantly reduce FEXT. It is the most effective way to suppress FEXT!

4.5 Dielectric Thickness

Dielectric thickness, to a large extent, affects crosstalk interference. In general, bringing the wiring layer closer to the power supply layer (Vcc or ground) can reduce crosstalk interference. The exact value of the improvement effect needs to be determined by simulation.

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Table 4-1 Example of Superposition of High-speed Material

Layer	Layer Description	Layer Design	Layer Thickness (mil)	er
Solder Mask			1	3.5
Layer 1	TOP	0.5oz+Plating	1.6	
Prepreg		1080HR(RC68%)	3.2	3.44
Layer 2	L2	1.0oz	1.2	
Core		0.1	4	3.72
Layer 3	L3	1.0oz	1.2	
Prepreg		2116MR(RC55%)*2	8.5	3.6
Layer 4	L4	1.0oz	1.2	
Core		0.1	4	3.72
Layer 5	L5	1.0oz	1.2	
Prepreg		2116MR(RC55%)*2	8.5	3.6
Layer 6	L6	1.0oz	1.2	
Core		0.1	4	3.72
Layer 7	L7	1.0oz	1.2	
Prepreg		2116MR(RC55%)*2	8.5	3.6
Layer 8	L8	1.0oz	1.2	
Core		0.1	4	3.72
Layer 9	L9	1.0oz	1.2	
Prepreg		1080HR(RC68%)	3.2	3.44
Layer 10	BOT	0.5oz+Plating	1.6	
Solder Mask			1	3.5
Theoretical Finished Layer Thickness :		1.6MM±10%		

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5 Supplies Requirements

You must design package and board with the integrity of the power distribution network (PDN) in mind. To ensure power supply integrity you must extract and simulate the power supplies.

5.1 Power supplies DC and AC Requirements

The power supply DC and AC voltage requirements are provided in Table 5-1 and Table 5-2.

Table 5-1 Power Supply Requirements for PI7C9X3G606GP-808GP-816GP-1632GP

Symbol	Description	Min.	Typ.	Max.	Unit
VDDC[2]	Digital Core Power	0.90	0.95	0.99	V
C_VDDR[2]	Referenc Clock Power	0.90	0.95	0.99	
VDDR[3]	Digital I/O Power	1.71	1.8	1.89	
C_VDDR[3]	Reference Clock Power	1.71	1.8	1.89	
VP[2]	PCI Express Analog Power	0.90	0.95	0.99	
VPH[3]	PCI Express Analog High Voltage Power	1.71	1.8	1.89	

Note:
2. Peak to peak AC noise < 5% of DC level.
3. Peak to peak AC noise < 3% of DC level.
4. If you use switching-mode power supply:
   - Use a ferrite bead or inductive choke to isolate supply noise and filter edge rate and amplitude at package pins
   - Keep switching frequency to 400 KHz or less to avoid supply tones in DJ sensitive region
5. For VPH analog power supply, use a low-dropout (LDO) voltage regulator to minimize the supply noise.

The power supply DC and AC voltage requirements are provided in Table 5-1 and Table 5-2.

Table 5-2 Power Supply Requirements for PI7C9X3G1224GP

Symbol	Description	Min.	Typ.	Max.	Unit
VDDC[2]	Digital Core Power	0.95		0.99	V
C_VDDR[2]	Referenc Clock Power	0.95		0.99	
VDDR[3]	Digital I/O Power	1.71	1.8	1.89	
C_VDDR[3]	Reference Clock Power	1.71	1.8	1.89	
VP[2]	PCI Express Analog Power	0.95		0.99	
VPH[3]	PCI Express Analog High Voltage Power	1.71	1.8	1.89	

Note:
2. Peak to peak AC noise < 5% of DC level.
3. Peak to peak AC noise < 3% of DC level.
4. If you use switching-mode power supply:
   - Use a ferrite bead or inductive choke to isolate supply noise and filter edge rate and amplitude at package pins
   - Keep switching frequency to 400 KHz or less to avoid supply tones in DJ sensitive region
5. For VPH analog power supply, use a low-dropout (LDO) voltage regulator to minimize the supply noise.

5.2 Package and Board Guidelines – Supply Routing

The power supplies should be routed with the following guidelines in mind:

- When possible, put power supply on power planes adjacent to GND reference to minimize PDN impedance.
- Ensure that SERDES power is well shielded from other noisy sourece on packaging and PCB to minimize any capacitive or inductive noise coupling.

The VP/VPH power supply is very sensitive to supply noise and care must be taken to reduce loop inductance in the package.

5.3 Package Guidelines – Inductance

Recommended maximum supply inductance for each of VP and VPH is given in Table 5-3. A power integrity simulation should be performed to ensure noise and low impedance requirements are met with the given package parasitics.

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Table 5-3 Maximum Supply inductance

Parameter	Value	Units	Notes
Maximim package inductance per PHY	0.5	nH	Measured from each chip connect (C4) cluster as a single port to the BGAs grouped as a single port
Masimum PC Board loop inductance per PHY	1.5	nH	Recommendation is that a single BGA can be up to 3nH, two BGAs up to 1.5nH. Achieving this low Lext typically requires a pin field capacitor of 100nF at each supply BGA to the closet return BGA

5.4 Board Bypass Capacitors

Supply bypass capacitors are recommended to minimize power supply noise. Simulation of the power delivery network is required to determine the actual values. Depending on their size, each capacitor will have a different equivalent series resistance (ESR) and equivalent series inductance (ESL) that will determine the given capacitor's effectiveness over a frequency range. In general, several low-value capacitors should be placed as close as possible to the package pins. Larger-value capacitors can be placed farther away.

Table 5-4 provieds an example list of typical capacitor values. Note that these example capacitor values may not be appropriate for your package and board environment. A power integrity analysis must be performed to ensure on-die supply noise requirement are met.

Table 5-4 Example Supply Bypass Capacitors

Component	Value[a]
Power Supply Bypass Capacitors	0.01uF, 0.1uF, 4.7uF, 10.0uF

The supply bypass capacitors should be connected as close as possible to the package pins to ensure a tight return path and maximize their effectiveness. When connecting from the package pin to the bypass capacitors use as wide a plane as possible to reduce both inductive as well as resistive losses. Typical capacitor placement can be under the package in BGA pin field using low inductance attachment, or placed close at the peripheral of BGA on the same side.

The key to get proper PCB decoupling is to have low mounting inductance with proper layout practice, have power plane and closely spaced GND plane for low spread inductance. Capacitor choices are made to suppress resonance in the low / mid frequency range and get as flat and low impedance curve as possible.

5.5 Board Guidelines – Ferrite Bead for Analog Power Supply

A ferrite bead (or inductive choke) is to be used to isolate power-supply noise presented on each on-board power supply. Power interity and noise analysis are recommended to determine the correct specification of the ferrite bead to be used. A sample ferrite bead specification is listed below (for example only).

Table 5-5 Example Ferrite Bead

Component	Impedance @ 100 MHz	Rated Current	DCR
EXCML45A910U	91 Ω	3A	16 mΩ

When using ferrite bead, put a bulk decoupling cap (for example, 10uF or 22uF) between the package pin and ferrite. This will help supply current at mid / low frequency as ferrite inductance may limit the current at low end.

5.6 Power Integrity Analysis

To enable power interity (PI) analysis of the power distribution network (PDN), current waveforms along with Cdie and Rdie are provided with the PHY deliverables. You should simulate the supply noise using this PI enablement along with a model of your package and board to ensure that on-die DC and AC supply requirements are being met.

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PDN impedance has to be low enough to support the amount of switching current. This can be checked by running a transient simulation with provide Icct of various power supply, with Rdie / Cdie and packaging PDN and board decoupling network.

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6 CLOCK Scheme

6.1 Reference Clock Input Pairs

The reference clock input pins connect to external 100MHz differential clock. The signal must match to LVPECL, LVDS or HCSL spec.

HCSL PCIe clock ref. clock drive can be AC or DC.

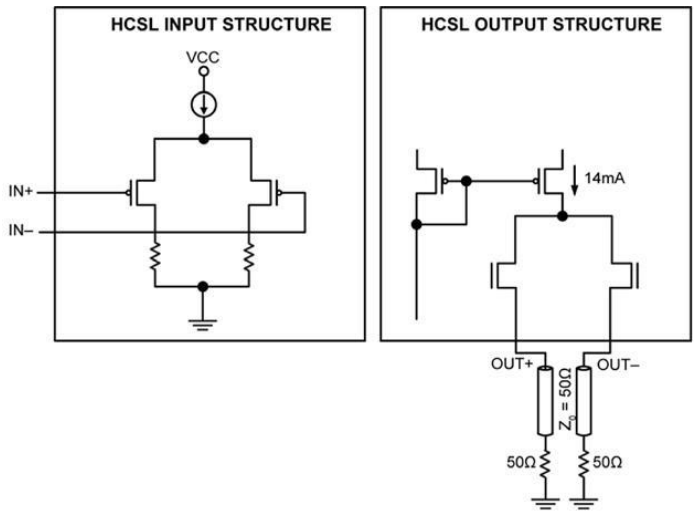
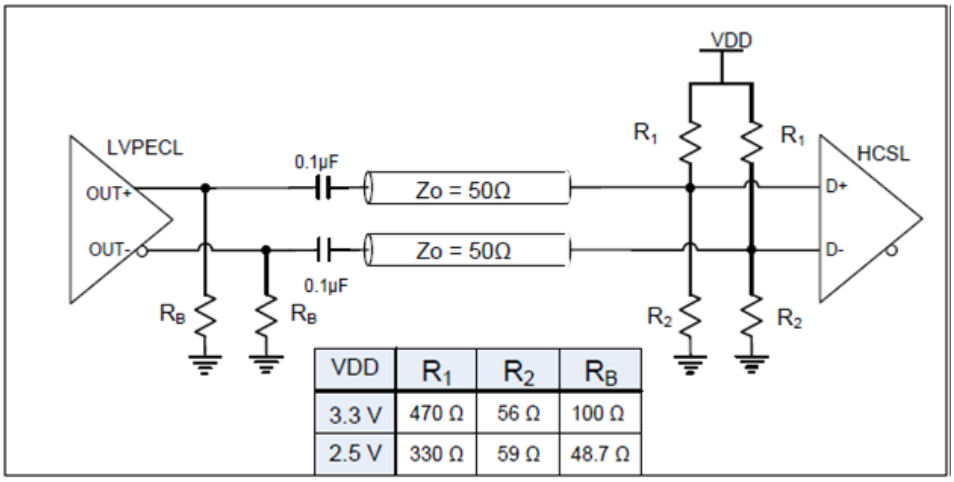


Figure 4 HCSL Input / Output Structure

LVPECL ref. clock use AC drive only.
The reference clock input pins need to add AC Coupling and level-shift circuits to rebuild common-mode voltage.



VDD	R_1	R_2	R_B
3.3 V	470 Ω	56 Ω	100 Ω
2.5 V	330 Ω	59 Ω	48.7 Ω

Figure 5 LVPECL to HCSL Interface

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LVDS ref. clock use DC drive, since LVDS V_swing has 50% smaller than HCSL and LVPECL. The reference clock input pins do not need to add AC Coupling. LVDS clock input with 1.25 Vcm, LVDS need connect 100ohm / 1% resistor between the reference clock input pins.

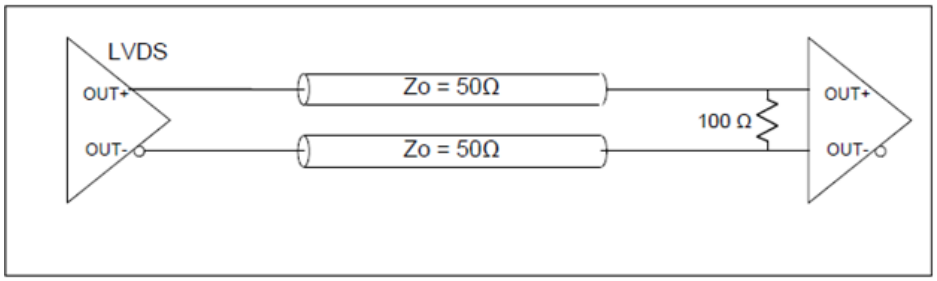
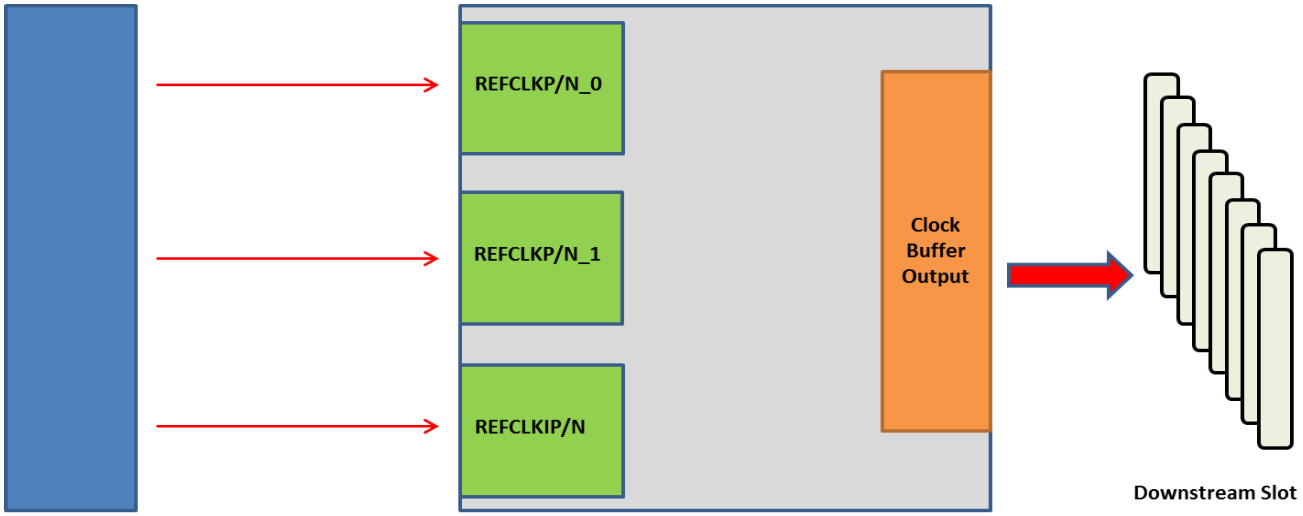


Figure 6 LVDS Single DC Termination Interface

A 100nF capacitor should be placed between the clock source and the packet switch. The purpose of this capacitor is to achieve AC coupling. This AC Coupling ensures the Packet Switch is compatible with the differential clock signals regardless the type of the clock. The input clock signals must be delivered to the clock buffer cell through an AC-coupled interface so that only the AC information of the clock is received, converted, and buffered.

6.2 Reference Clock Mode

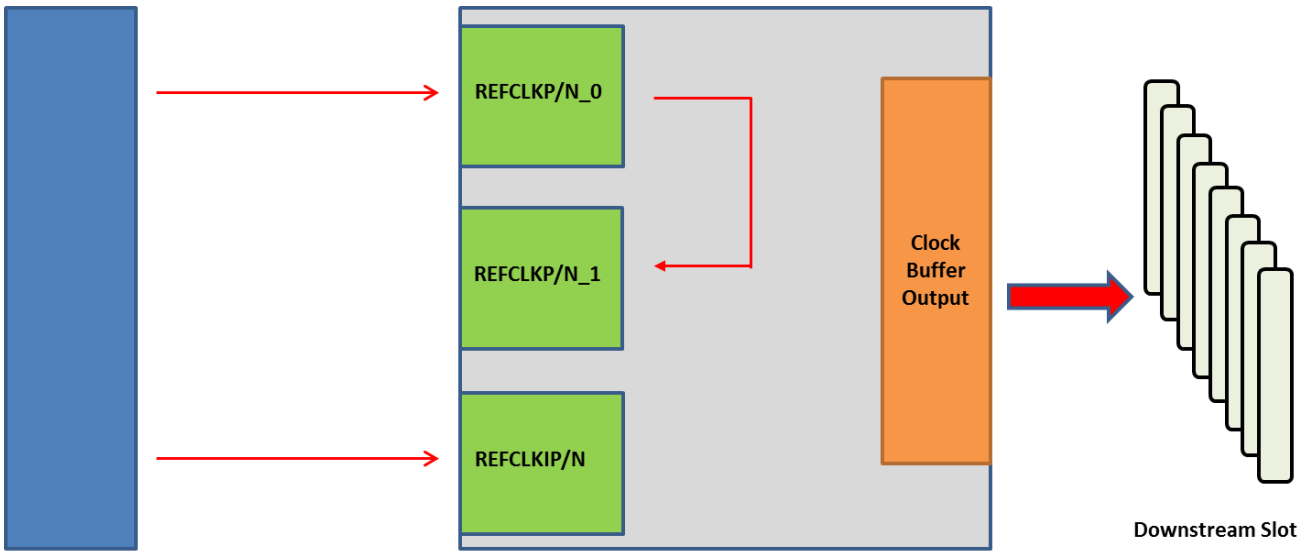
There is some block diagrams to indicate the CKMODE / CLKBUF_INPUT_SEL in circuit design.



Pin Name	Status	Note
CKMODE	Pull-up	BASE MODE
CLKBUF_INPUT_SEL	Pull-up	

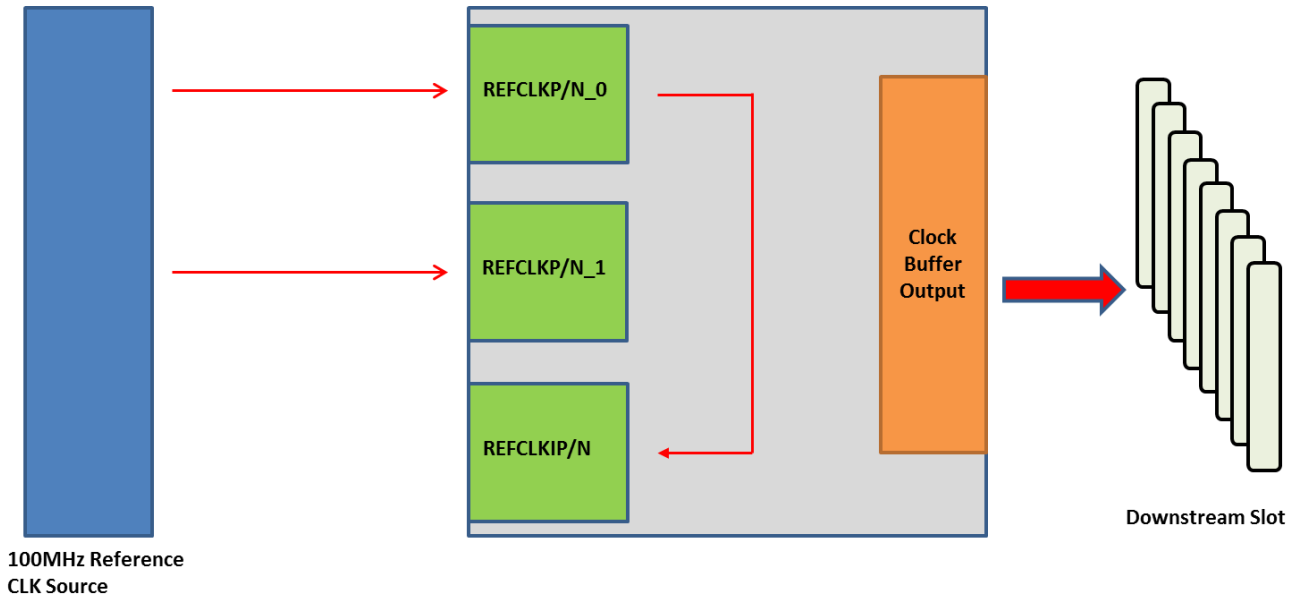
Figure 7 REFCLK Connections 1

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Pin Name	Status	Note
CKMODE	Pull-down	BASE MODE
CLKBUF_INPUT_SEL	Pull-up	

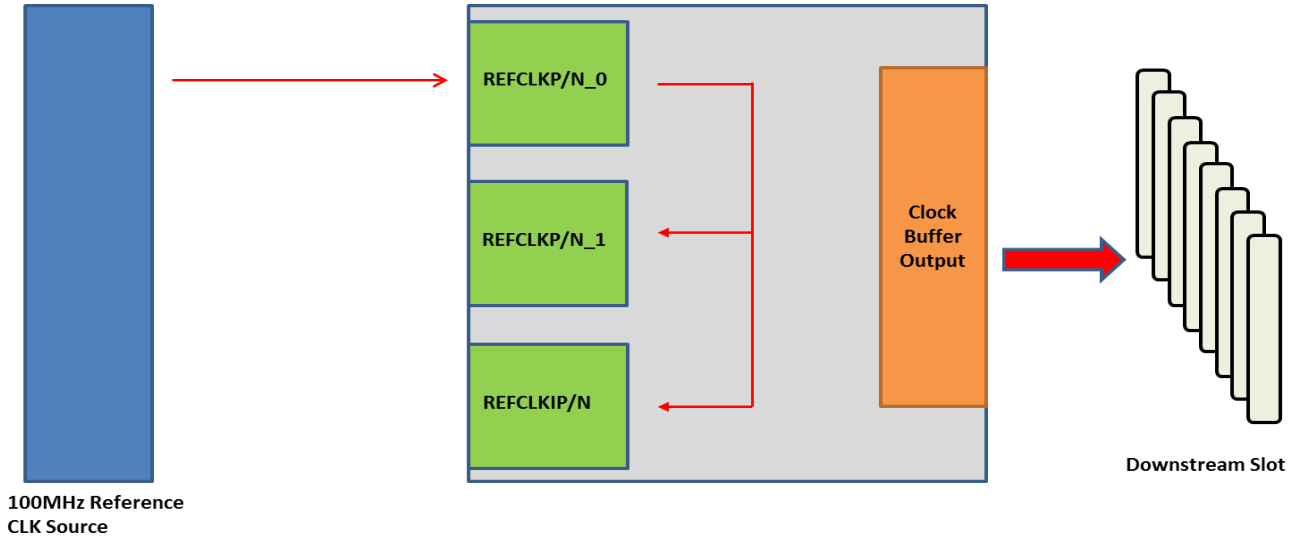
Figure 8 REFCLK Connections 2



Pin Name	Status	Note
CKMODE	Pull-up	BASE MODE
CLKBUF_INPUT_SEL	Pull-down	

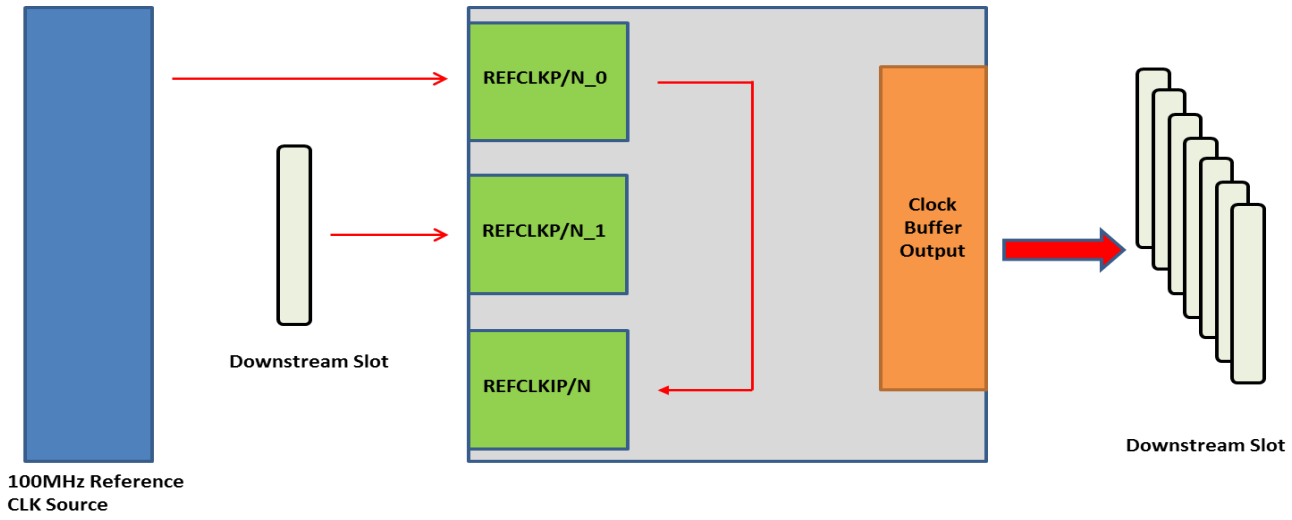
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Figure 9 REFCLK Connections 3



Pin Name	Status	Note
CKMODE	Pull-down	BASE MODE
CLKBUF_INPUT_SEL	Pull-down	

Figure 10 REFCLK Connections 4



Pin Name	Status	Note
CKMODE	Pull-up	SSC-Isolation
CLKBUF_INPUT_SEL	Pull-down	

Figure 11 REFCLK Connections 5

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Table 6-1 Clock Scheme with Gen3 Family Part Number

	PI7C9X3G606P	PI7C9X3G808GP	PI7C9X3G816GP	PI7C9X3G1632GP
Tile 0	REFCLKP/N_[1:0] / REFCLKIP/N / REFCLKOP/N_[7:4,1]	REFCLKP/N_[1:0] / REFCLKIP/N / REFCLKOP/N_[7:0]	REFCLKP/N_[2:0] / REFCLKIP/N / REFCLKOP/N_[7:0]	REFCLKP/N_0[2:0] / REFCLKIP/N_0 / REFCLKOP/N_0[7:0]
Tile 1	X	X	X	REFCLKP/N_1[2:0] / REFCLKIP/N_1 / REFCLKOP/N_1[7:0]

	PI7C9X3G1224GP
Tile 0	REFCLKP/N_0[2:0] / REFCLKIP/N_0 / REFCLKOP/N_0[7:0]
Tile 1	REFCLKP/N_1[1:0] / REFCLKIP/N_1 / REFCLKOP/N_1[5:4,1:0]

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7 GPIO Multi-Function Pins

The packet switch provides 32 GPIO pins each tile and can be configured as general GPIO pins, hot plug pins or L1.1 control pins. Hot plug pins and L1.1 strap pins are reset latch. After reset to "1" it will be released as GPIO.

7.1 General Purpose Inputs and Outputs

When both HOT_PLUG_ENx_L (x=0 or 1) and PM_L11_ENx_L (x=0 or 1) strap pins are pulled up, all GPIO_x (x=0 or 1) pins are configured as general purpose input and output function.

Offset 0x31Ch / 0x320h define the direction and source/destination bit for each GPIO. Below table is an example for GPIO[0] Source/Destination setting. For other GPIO, please see datasheet.

BIT	FUNCTION	TYPE	DESCRIPTION
0	GPIO[0] Source/Destination	RW	As Input: 0b: Input Data Register (offset 328h[0] 1b: General Interrupt (INTx, MSI or PEX_INTA#) As Output: 0b: From GPIO[0] Output Data register (offset 330h[0]) 1b: Reserved
1	GPIO[0] Direction Control	RW	0: Input 1: Output

Offset 0x328h / 0x32Ch define the input data register for each GPIO. Below table is an example for GPIO[0] Input Data setting. For other GPIO, please see datasheet.

BIT	FUNCTION	TYPE	DESCRIPTION
0	GPIO[0] Input Data	RO	GPIO[0] Input Data Return 0 if GPIO[0] is configured as an output (offset 31Ch[1]=1) Return the state of GPIO[0] pin if GPIO[0] is configured as an input (offset 31Ch[1]=0)
1	GPIO[1] Input Data	RO	GPIO[1] Input Data Return 0 if GPIO[1] is configured as an output (offset 31Ch[3]=1) Return the state of GPIO[1] pin if GPIO[1] is configured as an input (offset 31Ch[3]=0)

Offset 0x330h / 0x334h define the output data register for each GPIO. Below table is an example for GPIO[0] Output Data setting. For other GPIO, please see datasheet.

BIT	FUNCTION	TYPE	DESCRIPTION
0	GPIO[0] Output Data	RW	GPIO[0] Output Data The value written to this bit is driven to GPIO[0] output if GPIO[0] is configured as an output (offset 31Ch[1]=1)
1	GPIO[1] Output Data	RW	GPIO[1] Output Data The value written to this bit is driven to GPIO[1] output if GPIO[1] is configured as an output (offset 31Ch[3]=1)

Example: GPIO[3:0] are configured to high level outputs, GPIO[7:4] are configured to inputs.

a. Set 0x31Ch, bit[15:0] as 0x00AAh. It can able to progeam into EEPROM/SMBUS/I2C or through CFG directly.
b. Set 0x330h, bit[3:0] as 0xFh. It means GPIO[3:0] will output high level based on 0x330h, bit[3:0] value.
c. Read 0x328h, bit [7:4] value to get current GPIO[7:4] setting as they are inputs.

Table 7-1 GPIO with Gen3 Family Part Number

	PI7C9X3G606P	PI7C9X3G808GP	PI7C9X3G816GP	PI7C9X3G1632GP	PI7C9X3G122GP
Tile 0	GPIO[23:17,15:12,9:0]	GPIO[31:0]	GPIO[31:0]	GPIO_0[31:0]	GPIO_0[31:0]
Tile 1	X	X	X	GPIO_1[31:0]	GPIO_1[26,19:16,3:0]

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7.2 Managed Hot Plug Signals

When both HOT_PLUG_ENx_L and SURPRISE_HP_x strapping pins are pulled down, some GPIO pins are configured as managed hot plug pins.

GPIO_x[31,30,26,23:16,7:0] (x=0 or 1) are configured as general GPIO pins
GPIO_x[29:27,25,24,15:8] (x=0 or 1) are configured as hot plug pins

Table 7-2 Managed Hot Plug with Gen3 Family Part Number

	PI7C9X3G606P	PI7C9X3G808GP	PI7C9X3G816GP	PI7C9X3G1632GP	PI7C9X3G1224GP
Tile 0	Hot Plug_EN_L	Hot Plug_EN_L	Hot Plug_EN_L	Hot Plug_EN0_L	Hot Plug_EN0_L
Tile 1	X	X	X	Hot Plug_EN1_L	Hot Plug_EN1_L

How to enable Managed Hot Plug:

1. IO Expender and circuit is required
2. HOT_PLUG_ENx_L and SURPRISE_HP_x are pull-down
3. PDC_[x]_L pull-down
4. Set "Slot Capabilities register" 0x7Ch,[6:5] (bit[5]: Hot-Plug Surprise, bit[6]: Hot-Plug Capable) as 10b in EEPROM

Make sure Managed Hot Plug enable / disable in cfg register

1. Check downstream port "Slot Capabilities register" 0x7Ch,[6:5] (bit[5]: Hot-Plug Surprise, bit[6]: Hot-Plug Capable) as 10b.
2. Check Upstream port "DEVICE CONFIGURATION 2 REGISTER" 0x50Ch,[2:1] (bit[1]: HotPlug_Enable, bit[2]: Surprise_Hot_Plug) as 01b
3. Check "OPERATION MODE REGISTER" 0x348h,bit[24:23] (bit[23]: hotplug_pin_en_l, bit[24]: surprise_hp_en) as 01b

7.3 Surprise Hot Plug Signals

When HOT_PLUG_ENx_L strapping pin is pulled down and SURPRISE_HP_x strapping pin is pulled up, some GPIO pins are configured as surprise hot plug pins.

GPIO[31:24,15:8] are configured as general GPIO pins
GPIO[23:16,7:0] are configured as surprise hot plug pins

Table 7-3 Surprise Hot Plug with Gen3 Family Part Number

	PI7C9X3G606P	PI7C9X3G808GP	PI7C9X3G816GP	PI7C9X3G1632GP	PI7C9X3G1224GP
Tile 0	EEPROM	SURPRISE_HP	SURPRISE_HP	SURPRISE_HP_0	SURPRISE_HP_0
Tile 1	X	X	X	SURPRISE_HP_1	X

How to enable Surprise Hot Plug:

1. Pull-down HOT_PLUG_EN_L strap pin
2. Pull-up SURPRISE_HP strap pin (3G606GP doesn`t support this pin)
3. PDC_[x]_L connect to PRSNT pin of PCI Express slot and pull-up
4. Set "Slot Capabilities register" 0x7Ch,[6:5] (bit[5]: Hot-Plug Surprise, bit[6]: Hot-Plug Capable) as 11b in EEPROM

Make sure Surprise Hot Plug enable / disable in cfg register

1. Check downstream port "Slot Capabilities register" 0x7Ch,[6:5] (bit[5]: Hot-Plug Surprise, bit[6]: Hot-Plug Capable) as 11b.
2. Check Upstream port "DEVICE CONFIGURATION 2 REGISTER" 0x50Ch,[2:1] (bit[1]: HotPlug_Enable, bit[2]: Surprise_Hot_Plug) as 11b

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3. Check "OPERATION MODE REGISTER" 0x348h,bit[24:23] (bit[23]: hotplug_pin_en_l, bit[24]: surprise_hp_en) as 11b

7.4 PM L1.1 Control Signals

When PM_L11_ENx_L is pulled down, PM L1.1 function is enabled, and some GPIO pins are configured as PM L1.1 control signals.

GPIO[31:16,7:0] are configured as general GPIO pins
GPIO[15:8] are configured as PM L1.1 control pins

Table 7-3 PM L1.1 Control with Gen3 Family Part Number

	PI7C9X3G606P	PI7C9X3G808GP	PI7C9X3G816GP	PI7C9X3G1632GP	PI7C9X3G1224GP
Tile 0	PM_L11_EN_L	PM_L11_EN_L	PM_L11_EN_L	PM_L11_EN0_L	PM_L11_EN0_L
Tile 1	X	X	X	PM_L11_EN1_L	X

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8 SMBUS/I2C

The packet switches support the SMBus/I2C function. The SMBus/I2C address can be configured by I2C_ADDR[2:0]. The address of Tile_0 is 1101XXX_b (Tile_1 is address of Tile_0 [bit 0] +1, example: If Tile_0 address configured as D0h, Tile_1 address would be D2h). Bit 0 is used to indicate R/W. Bit 1, 2, and 3 corresponds to I2C_ADDR [2], I2C_ADDR [1], and I2C_ADDR [0]. Please refer to datasheet to get more detail information.

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9 Thermal Sensor Access Guidelines

Each Tile includes three thermal sensors and below three reigsters are used to access the thermal sensors.

Thermal Sensor 0 – access cfg 0x5d8
Thermal Sensor 1 – access cfg 0x5dc
Thermal Sensor 2 – access cfg 0x5e0
Bit[1]: Thermal Sensor Single Run
Bit[19:8]: Thermal Sensor Data
Bit[23]: Thermal Sensor Data Done

How to Get Thermal Sensor 0 tempature?
Wr 0x5d8 00000002
Rd 0x5d8 00882200
-----date done =1
-----Ts data = ‘h822

Using translate formula to get centigrade degrees.
K=79.925;
Y=237.7;
Hex to Dec -----‘h822 -→ ‘d 2082
TEMP = (N/4094)*Y –K = (2082/4094)*237.7-79.925
=40.95(degrees in centigrade);

Table 9-1 Thermal Sensor Access with Gen3 Family Part Number

	PI7C9X3G606P	PI7C9X3G808GP	PI7C9X3G816GP	PI7C9X3G1632GP	PI7C9X3G1224GP
Tile 0	V	V	V	V	V
Tile 1	X	X	X	V	V

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10 Serial Loopback Test Mode

The switch implemets a serail loopback test mode which can be used to test signal integrity.

Please follow below procedures to enter serial loopback test mode.

a. Make sure the link is up.
   If want to test up port, up port need link to root complex.
   If want to test down port, down port need link to the related down port device.

b. Set Target Link Speed
   Write offset 98h.bit[3:0] (Link Control 2 Register.bit[3:0]: target link speed)
   - Gen1: 0001b
   - Gen2: 0010b
   - Gen3: 0011b

c. Retrain the link
   Write 78h.bit[5] for down ports
   If the test port is up port, need set root complex retrain bit (Link Control Register.bit[5], retrain)

d. Set switch as the loopback master
   Write offset 390h.bit[11] to 1 to set as loopback master

e. Check Error Counter by SMBUS/I2C interface
   Write offset 434h.bit[31] to enable loopback data
   Read offset 434h.bit[14] to get lookback data
   Read offset 434h.bit[15] to get right loopback data
   Check Error Counter from offset 434h.bit[27:16]
   (If error counter is 0, it indicates the signal integrity is good. Otherwise, SI may have problem.)

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11 IREF (RESREF)

Electronic circuits in semiconductor processes, especially for integrated circuits made from analog or mixed-signal circuits, usually require a reference current or voltage. A simple current reference circuit of prior art. Metal oxide semiconductor field effect transistors are coupled to each other to form a current mirror. Resistance R(In this case, it's R149.) is used to form a current reference circuit in one of the branches of the current mirror, thus producing a reference current. One end of resistance R is grounded and the other is connected to the pin of the IC. And the closer it is to the IC , the better.

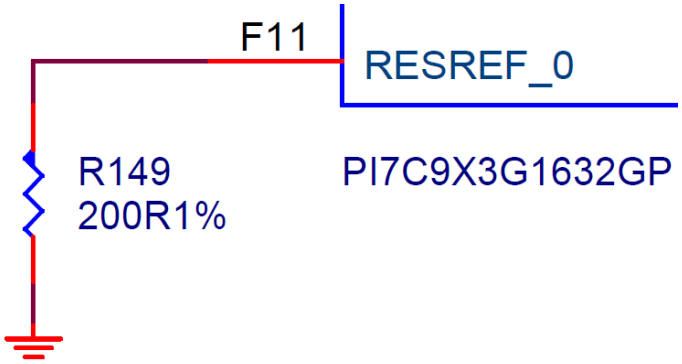


Figure 12 IREF Resistance

Table 11-1 Thermal Sensor Access with Gen3 Family Part Number

	PI7C9X3G606P	PI7C9X3G808GP	PI7C9X3G816GP	PI7C9X3G1632GP	PI7C9X3G1224GP
Tile 0	RESREF	RESREF	RESREF	RESREF_0	RESREF_0
Tile 1	X	X	X	RESREF_1	RESREF_1

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12 Serial Hot Plug (SHP)

With external IO expanders and Hot-Plug controllers, Diodes Gen3 Packet Switch supports Serial Hot-Plug function for each port. By default, the Serial Hot-Plug circuit is enabled. The block diagram below shows how the Diodes Gen3 Packet Switch implement the control circuit of the Serial Hot-Plug with external IO expanders.

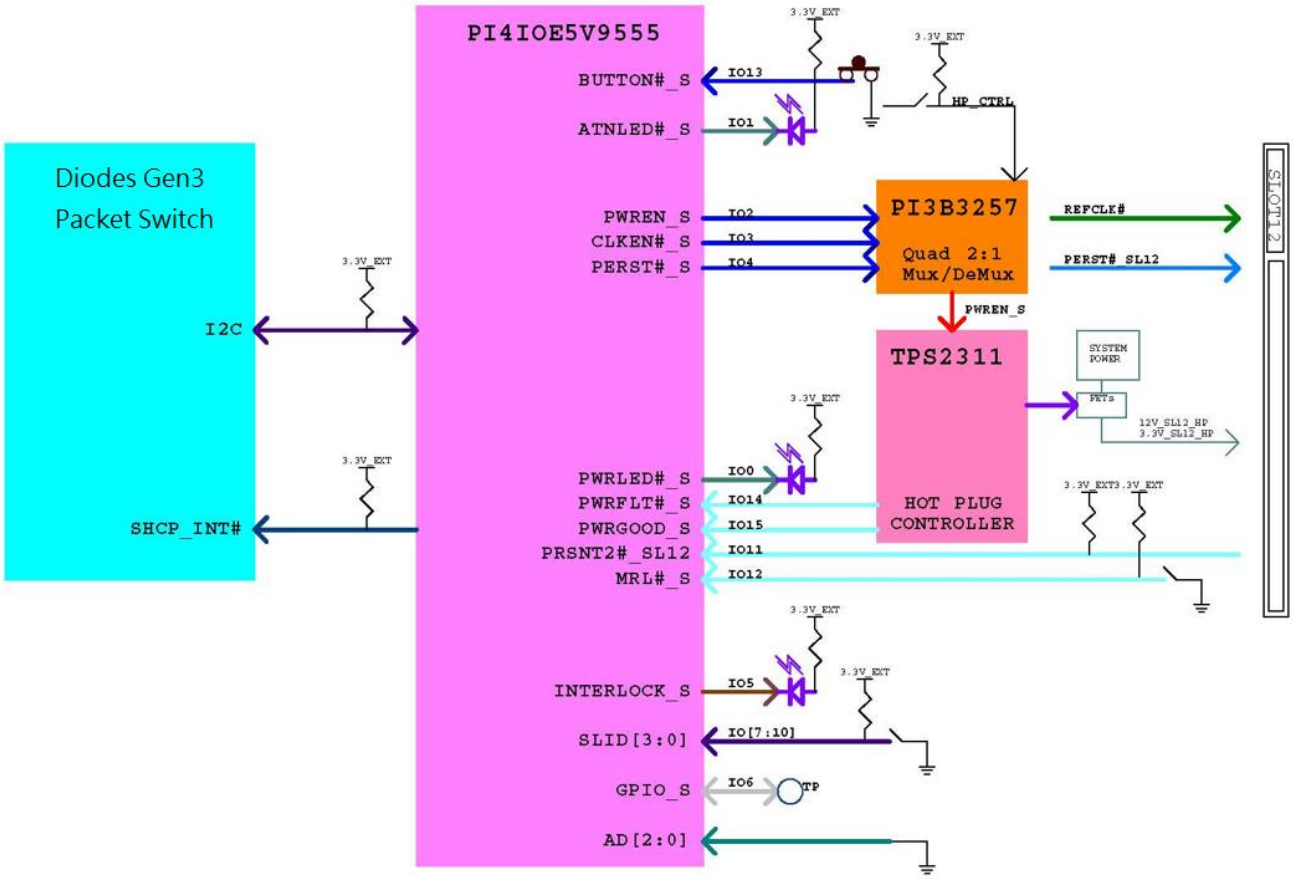


Figure 13 Serial Hot Plug Circuits

As shown in Figure 13 Serial Hot Plug Circuits

above, the Serial Hot-Plug controller consists of PI4IOE5V9555 I/O expander, a TI TPS2311 dual hot-swap power controller, a PI3B3257 2-to-1 multi-plexer, two power MOSFET DMN3010LSS-13, LEDs, manual switch, dipswitches and resistors. The Gen3 Packet Switch master I2C interface is designed for the specific control use of the Serial Hot-Plug controller. The master I2C interface connected to the I/O expander and the interrupt output from the I/O expander connects to the SHPCINT_L of the Gen3 Packet Switch. When power is applied to the Gen3 Packet Switch, the master I2C interface will scan the bus and attempts to detect the presence of the I/O expander. If an I/O expander is detected, the I2C master will program it as a “remote parallel Hot-Plug controller” and assign an available Serial Hot-Plug port to the I/O expander. The IO expender then generates PWREN_S, CLKEN#_S, and PERST#_S outputs to control the power enable, CLKEN#_S, and PERST# of the SHP SLOT and drives the LEDs at PWRLED#_S, ATNLED#_S and INTERLOCK#_S outputs. Also, it accepts BUTTON#_S, PWRFLT#_S, PWRGOOD_S, PRSNT2# and MRL#_S inputs and generates interrupts to Gen3 Packet Switch for Serial Hot-Plug status changes.

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Remember to provides dip switch for setting the SLOTID [3:0] for the SLOT, dip switch to control the Serial Hot-Plug controller functions (enable/disable it), dip switch to set ADD [2:0] of the IO expander, and a test point for access the GPIO pin.

If you want to implement the SHP function, in addition to setting the SLOT Capability register value of the corresponding slot, you also need to set the relevant SHP strap pins of the Gen3 Packet Switch. In addition, please make sure that your platform can support SHP：

1) HOT_PLUG_EN_L need to set to Low
2) IOE_ADDR[2:0] should be set related to SHP port
3) SURPRISE_HP_EN should be set to low to close surprise hot plug function if use Serial Hot Plug function

Table 12-1 Serial Hot Plug Port / IOE_ADDR[2:0] Mapping

Serial Hot Plug PORT	IOE_ADDR[2:0]
1	000
2	001
3	010
4	011
5	100
6	101
7	110

Table 12-2 Serial Hot Plug Application Step

Strap pin setting	Action & Step
MRL#_S：Off → On	There is no card in the slot as the system starts, and the card should be inserted at this time 1. MRL initially Off, while slot is empty 2. System power on, and enter into OS 3. Insert the card, and set MRL to On 4. Press Attention Button, PWR LED blink, OS start to power on the Slot 5. Eventually, PWR LED on, OS finish enumerate card
MRL#_S：On → Off	In system, use ATT button to remove the card 1. Press Attention Button, PWR LED should blink, system start power off slot 2. System eventually shows “card could be removed”, PWR LED should be off 3. Make sure slot power is off, remove the card 4. MRL set to Off

Table 12-3 Serial Hot Plug with Gen3 Family Part Number

	PI7C9X3G606P	PI7C9X3G808GP	PI7C9X3G816GP	PI7C9X3G1632GP	PI7C9X3G1224GP
Tile 0	V	V	V	V	V
Tile 1	X	X	X	V	V

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13 JTAG Pins

NAME	TYPE	DESCRIPTION
TCK	I	**Test Clock:** Used to clock state information and data into and out of the chip during boundary scan. When JTAG boundary scan function is not implemented, this pin should be left open (NC).
TDI	I	**Test Data Input:** Used (in conjunction with TCK) to shift data and instructions into the TAP in a serial bit stream. When JTAG boundary scan function is not implemented, this pin should be left open (NC).
TDO_0 / TDO	O	**Test Data Output:** Used (in conjunction with TCK) to shift data out of the Test Access Port (TAP) in a serial bit stream. When JTAG boundary scan function is not implemented, this pin should be left open (NC). For PI7C9X3G1632GP-3G1224GP, there are two test data output pins, TDO_0 and TDO_1. For PI7C9X3G816/808/606GP, there is only one test data ouput pin.
TDO_1	O	**Test Data Output:** Used (in conjunction with TCK) to shift data out of the Test Access Port (TAP) in a serial bit stream. When JTAG_SEL_L set high, this pin should be left open (NC). This pin is implemented by PI7C9X3G1632GP-3G1224GP only.
TMS	I	**Test Mode Select:** Used to control the state of the Test Access Port controller. When JTAG boundary scan function is not implemented, this pin should be pulled low through a 330-Ohm pull-down resistor.
TRST_L	I	**Test Reset (Active LOW):** Active LOW signal to reset the TAP controller into an initialized state. When JTAG boundary scan function is not implemented, this pin should be pulled low through a 330-Ohm pull-down resistor.
JTAG_SEL_L	I	**JTAG Selection (Active LOW):** When set high, JTAG pins used for controlling switch's boundary scan, when set low, JTAG pins used for accessing PCIe PHY internal registers.

Table 13-1 JTAG Pins Setting

PIN	Normal Operation	Switch Boundary Scan	PHY JTAG
JTAG_SEL_L	1	1	0
CHIPMODE[1:0]	00	10	00->10[Note 2]
TRST_L	0	1	1
TDO_0 / TDO	No Connection	JTAG Output	JTAG Output for Tile 0
TDO_1[Note]	No Connection	No Connection	JTAG Output for Tile 1

Note 1: TDO_1 is implemented by PI7C9X3G1224GP and PI7C9X3G1632GP only.

Note 2: CHIPMODE[1]=0 before PERSRT_L is de-asserted. Then, change CHIPMODE[1]=1 after PERST_L is de-asserted.

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14 PHY TX/RX Signal Adjustment

Although Diodes' Gen3 Packet Switch can automatically adjust the best connection parameters according to downstream components and environment conditions. Diodes' Gen3 Packet Switch still allows users freely and directly adjust some parameters.

Below is the part of Diodes' Gen3 Packet Switch that allows users to adjust：

14.1 PHY TX Peak-to-Peak Swing Adjustment

The Peak-to-Peak output swing of the TX, Vswing, is nominally equal to the power supply. However, the TX supports the boost function, which is enabled by the input signal of TX_VBOOST_LVL_EN.

TX_VBOOST_VREF is an internal voltage that determines the maximum possible swing and is determined by TX_VBOOST_LVL, while TX_IBOOST_LVL provides fine control of the boosted amplitude.

14.1.1 VBOOST

By entering CR interface, the offset 0x0Ch of PHY register is used for roughly tuning the VBOOST.
Following is the related bit：
Bit[9] ：-TX_VBOOST_LVL_EN (Enable override value for TX_VBOOST_LVL)
Bit[8:6] ：-TX_VBOOST_LVL (Override value for TX_VBOOST_LVL)
Note：The change of VBOOST register is globally, it would affect Gen1/Gen2/Gen3 TX Peak to Peak Swing simultaneously

Following Table lists typical maximum TX swing values under the listed conditions：

1) DUT：Diodes 3G808GP with CBB/CLB
2) Measured Port：
   A. Port 4 (SPHY2 / Lane0)
   B. Port 0 (SPHY0 / Lane0) (Gen3 only)
3) IBOOST, Offset 0x1N02h, N=Lane, when Offset 1002h = 5078 (value equals to IBOOST[3:0]='1111')

Program Data	TX_VBOOST_LVL_EN	TX_VBOOST_LVL	TX diff p-p Swing (mV)						
			Gen1_P4	Gen1_3.5dB	Gen2_P4	Gen2_3.5dB	Gen2_6dB	Gen3_P0	Gen3_P4
01D0 (Default)	1		984	866	947	716	700	573	992
0200	1	3'b000	942	824	903	680	664	545	942
0240	1	3'b001	942	824	903	680	664	545	942
0280	1	3'b010	942	824	903	680	664	545	942
02C0	1	3'b011	942	824	903	680	664	545	942
0300	1	3'b100	984	864	947	716	700	573	992
0340	1	3'b101	1121	990	1078	820	804	655	1130
0380	1	3'b110	1240	1080	1191	900	883	726	1246
03C0	1	3'b111	1304	1132	1252	946	926	770	1300

14.1.2 IBOOST

By entering CR interface, the offset 0x1N02h of PHY register is used for fine tuning the IBOOST (N：Lane number 0~3)
Following is the related bit：
Bit[8] ：- EN (Enable override values for inputs controlled by this register)
Bit[7] ：-VBOOST_EN (Override value for TX_VBOOST_EN)
Bit[6:3] ：-IBOOST_LVL (Override value for TX_IBOOST_LVL)
Note：The change of IBOOST register is by lane

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Following Table lists typical maximum TX swing values under the listed conditions：

1) DUT：Diodes 3G808GP with CBB/CLB
2) Measured Port：
   A. Port 4 (SPHY2 / Lane0)
   B. Port 0 (SPHY0 / Lane0) (Gen3 only)
3) VBOOST, Offset 0x0Ch, when 0x0Ch=01D0 (value equals to VBOOST[2:0]='100' and Vptx = 0.95V)

Program Data	IBOOST_LVL	TX diff p-p Swing (mV)						
		Gen1_P4	Gen1_3.5dB	Gen2_P4	Gen2_3.5dB	Gen2_6dB	Gen3_P0	Gen3_P4
5078 (Default)		986	866	946	718	701	574	993
5180	4'b0000	944	828	906	683	668	548	946
5188	4'b0001	948	830					
5190	4'b0010	950	832					
5198	4'b0011	952	835	914				
51A0	4'b0100	956	838		693	677	555	960
51A8	4'b0101	958	841					
51B0	4'b0110	961	843					
51B8	4'b0111	963	845	924				
51C0	4'b1000	966	848		702	686	563	974
51C8	4'b1001	969	851					
51D0	4'b1010	972	853					
51D8	4'b1011	975	855	935				
51E0	4'b1100	977	858		710	694	570	984
51E8	4'b1101	980	861					
51F0	4'b1110	983	863					
51F8	4'b1111	986	866		718	701	574	993

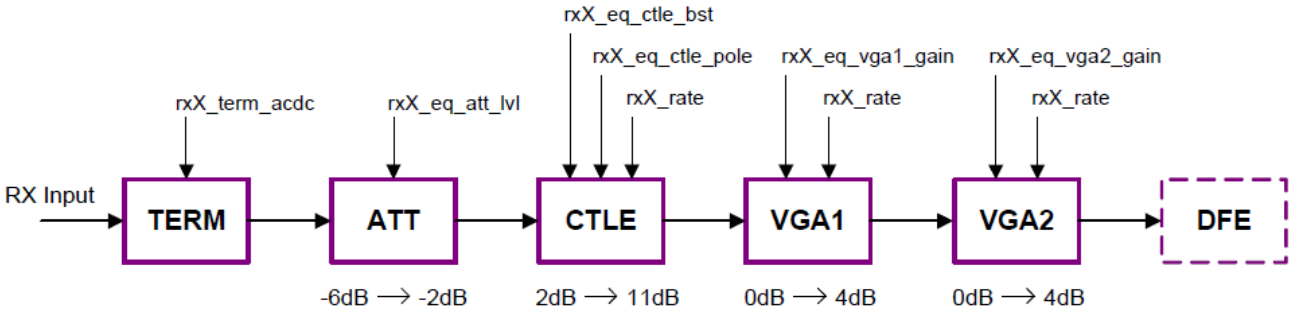
14.2 RX Eye Adjustment

On the receive (RX) side, receiver characteristics, such as equalization and termination, are user-controllable. A continuous-time linear equalizer (CTLE) is first used to compensate for channel loss. A decision feedback equalizer (DFE) is also implemented after CTLE to optimize the data eye further for high-speed data that exceeds ~6 Gbps.

14.2.1 RX Analog Front End

The analog front end (AFE) circuity receives off-ship data and boosts high-frequency components to compensate part of the channel loss with programmable or adaptive CTLE

Following figure shows RX AFE (Analog Front End) Diagram：



Termination (TERM)：this block is tuned by the PHY using an off-cip reference resistor, which is a single ended 50Ω termination to ground on each of RXP and RXM

Attenuator (ATT)：This block is responsible for attenuating the incoming signal to acceptable levels for the subsequent CTLE block to maintain linearity in case of a short-reach link

Continuous time-linear equalizer (CTLE)：This block provides high-frequency boost range from 2dB to 11dB. The CTLE block has programmable boost level, boost pole frequency, and boost BW control.

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Voltage gain amplifiers (VGA1 and VGA2)：The VGAs are implemented to maintain a constant signal swing. Voltage gain amplifiers (VGA1and VGA2) control analog gain, and bandwidth. There are two VGAs in series to improve the gain control range.

To achieve optimal link performance, the receiver equalization parameters are automatically tuned to their optimal values, but Diodes' Gen3 Packet Switch still provide related register for user to adjust：

AFE related register

By entering CR interface, following is the related register (Read only)：

1N13：N=Lane. Default value is 2007 (3G808GP on platform #47)

- Bit[15:11]：EQ_CTLE_BOOST
- Bit[10:7]：VGA2_GAIN
- Bit[6:3]：VGA1_GAIN
- Bit[2:0]：ATT_LVL

1N14：N=Lane. Default value is 00A2 (3G808GP on platform #47)

- Bit[2:0]：EQ_CTLE_POLE (suggest not to modify)

And following is R/W register for modification：

1Na9：N=Lane

- Bit[1]：RX_ANA_AFE_EN. Override value for RX_ANA_AFE_EN

1Nb0：N=Lane

- Bit[11]：RX_AFE_OVRD_EN. Override enable for AFE control
- Bit[10:7]：RX_ANA_AFE_VGA2_GAIN. Value for RX_ANA_AFE_VGA2_GAIN[3:0]
- Bit[6:3]：RX_ANA_AFE_VGA1_GAIN. Value for RX_ANA_AFE_VGA1_GAIN[3:0]
- Bit[2:0]：RX_ANA_AFE_ATT_LVL. Value for RX_ANA_AFE_ATT_LVL[2:0]

1Nb1：N=Lane

- Bit[7:3]：RX_ANA_AFE_CTLE_BOOST. Value for RX_EQ_CTLE_BOOST[4:0]
- Bit[2:0]：RX_ANA_AFE_CTLE_POLE. Value for RX_EQ_CTLE_POLE[2:0]

1Nb7：N=Lane

- Bit[0]：RX_ANA_AFE_UPDATE_EN. Value for RX_ANA_AFE_UPDATE_EN. If AFE_UPDATE_SELF_CLEAR_DISABLE=0, then this bit is self-clearing (in other words, it only asserts for one cr_clk cycle)

Step to change related value：

1) Enable 1Na9 Bit[1]：RX_ANA_AFE_EN
2) Enable 1Nb0 Bit[11]：RX_AFE_OVRD_EN
3) Change different value in 1Nb0 for VGA2_GAIN[10:7] / VGA1_GAIN[6:3] / ATT_LVL[2:0].
4) Change different value in 1Nb1 for CTLE_BOOST[7:3], suggest CTLE_POLE[2:0] not to change
5) Enable 1Nb7 Bit[0] to update 2) & 3) change：RX_ANA_AFE_UPDATE_EN

14.2.2 RX DFE (Decision Feedback Equalizer)

1) The Post-VGA output is summered by a 5-tap, continuously adaptive Decision Feedback Equalizer (DFE)
2) The 5-tap DFE is a half-rate architecture with first tap unrolling
3) The DFE can be disabled through top-level signal or register control

DFE related register

By entering CR interface, following is the related register

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1Na8：N=Lane
- Bit[8]：RX_CTL_OVRD_EN. Enable override values for this register's output [8:0]
- Bit[5]：RX_ANA_DFE_TAPS_EN. Enable bit

1N0a：N=Lane
- Bit[11]：EQ_OVRD_EN. Enable override value for RX_EQ inputs
- Bit[10:3]：EQ_DFE_TAP1. Override value for RX_EQ_DFE_TAP1

1N18：N=Lane
- Bit[13:7]：EQ_DFE_TAP3. Override value for RX_EQ_DFE_TAP3
- Bit[6:0]：EQ_DFE_TAP2. Override value for RX_EQ_DFE_TAP2

1N19：N=Lane
- Bit[13:7]：EQ_DFE_TAP5. Override value for RX_EQ_DFE_TAP5
- Bit[6:0]：EQ_DFE_TAP4. Override value for RX_EQ_DFE_TAP4

Steps to change TAP2 ~ TAP5 setting：

1) Set 1Na8：Bit[8] = 1 / Bit[5] = 1, which is 0120 at the time
2) Set 1N0a：Bit[11] to 1, set 1N0a Bit [10:3] to suitable value
3) Set 1N18：Set TAP2 & TAP3 value
4) Set 1N19：Set TAP4 & TAP5 value
5) Set 1Na8 = 0000

14.3 To disable Adapt_DFE and Adapt_AFE to check RX eye at different preset

Dioedes Gen3 Packet Switch provide adaptive and configurable Decision Feedback Equalizer (DFE) for user to fine-tune, and also can be disable. Following shows RX eye result when disable Adapt_DFE & Adapt_AFE at different preset value.

Testing environment：3G808EVB on platform #48

Related Register：

1) Set Preset value：380h (LTSSM CSR 0 Register) by PCI-BUDDY
   By setting 380h to set each lane's eq_peset_uplimited and eq_preset_dnlimited

BIT	FUNCTION	TYPE	DESCRIPTION	EEPROM/ I2C-SMBUS	DEFAULT
3:0	eq_preset_uplimited_0	RW	Define EQ evaluate upper limiter range of preset. This value is defined per lane.	Yes	Ah
7:4	eq_preset_dnlimited_0	RW	Define EQ evaluate down limiter range of preset. This value is defined per lane.	Yes	5h
11:8	eq_preset_uplimited_1	RW	Define EQ evaluate upper limiter range of preset. This value is defined per lane.	Yes	Ah
15:12	eq_preset_dnlimited_1	RW	Define EQ evaluate down limiter range of preset. This value is defined per lane.	Yes	5h
19:16	eq_preset_uplimited_2	RW	Define EQ evaluate upper limiter range of preset. This value is defined per lane.	Yes	Ah
23:20	eq_preset_dnlimited_2	RW	Define EQ evaluate down limiter range of preset. This value is defined per lane.	Yes	5h
27:24	eq_preset_uplimited_3	RW	Define EQ evaluate upper limiter range of preset. This value is defined per lane.	Yes	Ah
31:28	eq_preset_dnlimited_3	RW	Define EQ evaluate down limiter range of preset. This value is defined per lane.	Yes	5h

2) Phy Register：By entering CR interface to access phy register
   1N05：Read / Write register for Bit[14:12]

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14	EN	R/W	**Description**: Enable override values for all inputs controlled by this register **Value After Reset**: 0x0 **Reset Mask**: 0x1
13	ADAPT_DFE_EN	R/W	**Description**: Override value for rx_adapt_dfe_en **Value After Reset**: 0x0 **Reset Mask**: 0x1
12	ADAPT_AFE_EN	R/W	**Description**: Override value for rx_adapt_afe_en **Value After Reset**: 0x0 **Reset Mask**: 0x1

1N11：Read only register for Bit[13:12]

13	ADAPT_DFE_EN	R	**Description**: Value from ASIC for rx_adapt_dfe_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Reset Mask**: 0x0 **Volatile**: true
12	ADAPT_AFE_EN	R	**Description**: Value from ASIC for rx_adapt_afe_en (volatile) (actual reset value is 0xX) **Value After Reset**: 0x0 **Reset Mask**: 0x0 **Volatile**: true

Step：N = Lane no

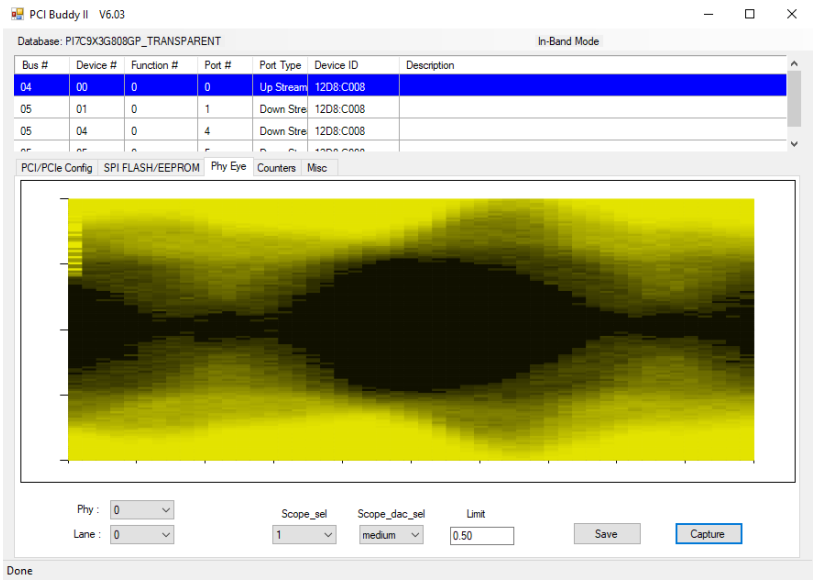
By entering CR interface

a) Check 1N11 value (for 3G808 on #49 is 7404h)
b) Write 1N05 = 7404
c) Write 3N08 = 1000
d) Write 3N08 = 1800
e) Write 3N10 = 0000
f) Read 1N11 (should be 440C)
g) Write 1N05 = 440C
h) Read 1N11 (should be 4404)
i) Check RX eye

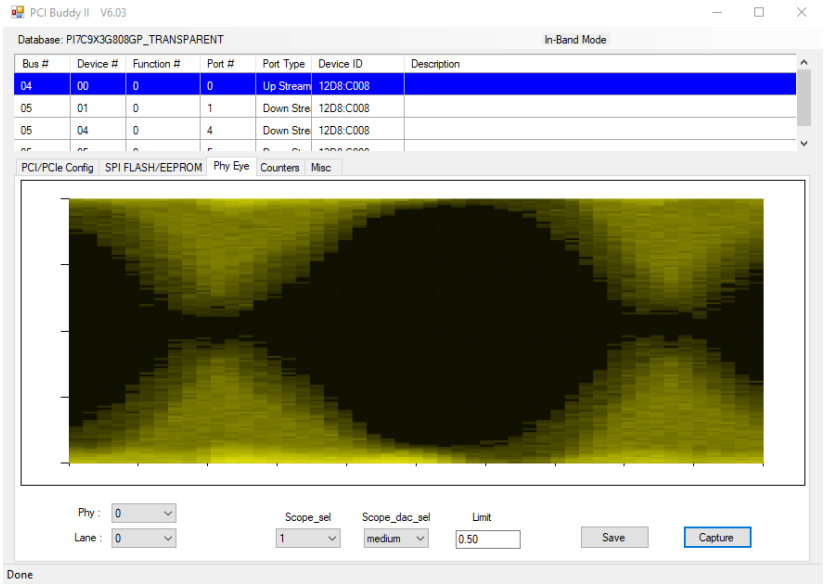
Check Result：

1) 380h = 11111111 (check by 3C8h)
   1N11 = 7404h (Enable ADAPT_AFE & ADAPT_DFE)

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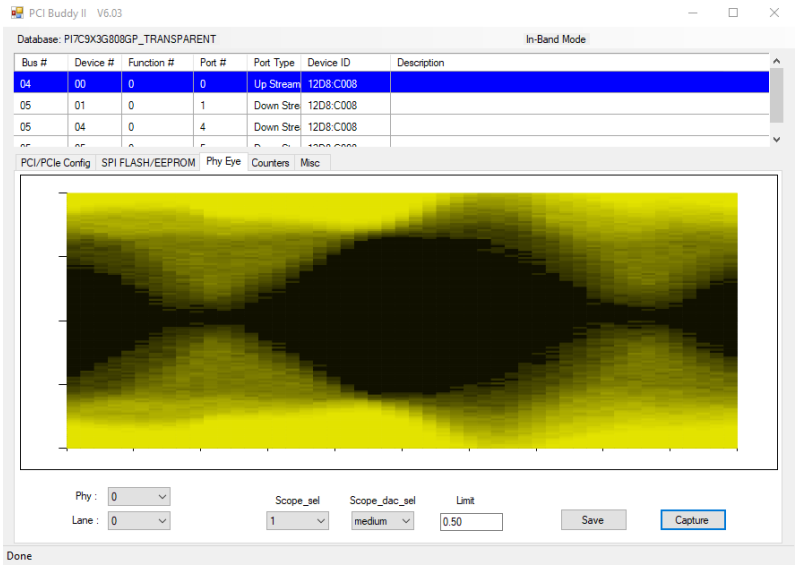


1N11 = 4404 (Disable ADAPT_AFE & ADAPT_DFE)

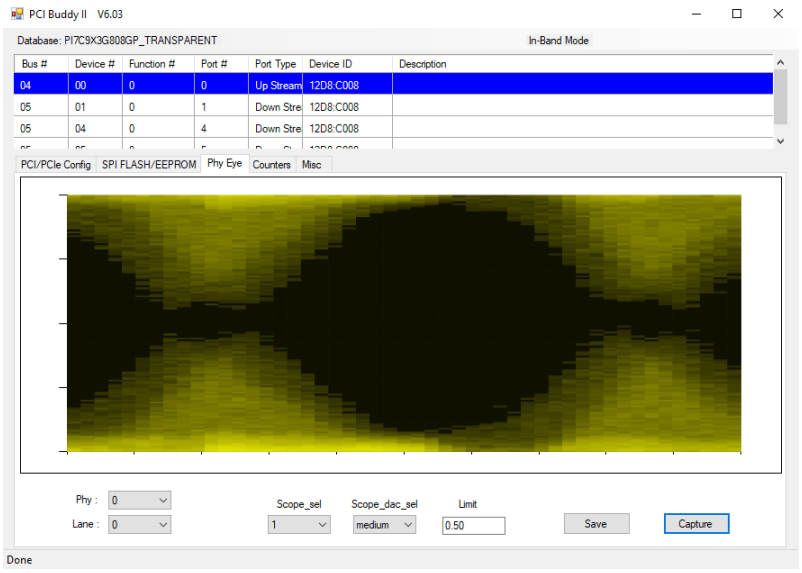


2) 380h = 22222222 (check by 3C8h)
1N11 = 7404h (Enable ADAPT_AFE & ADAPT_DFE)

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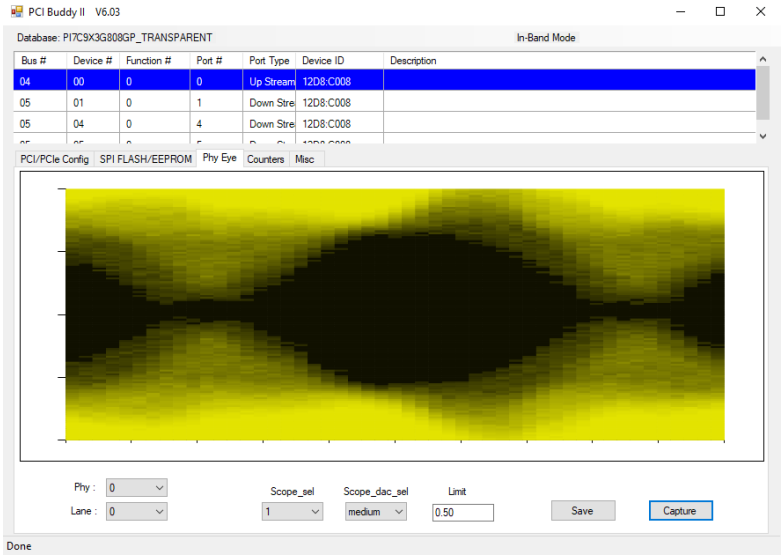


1N11 = 4404 (Disable ADAPT_AFE & ADAPT_DFE)

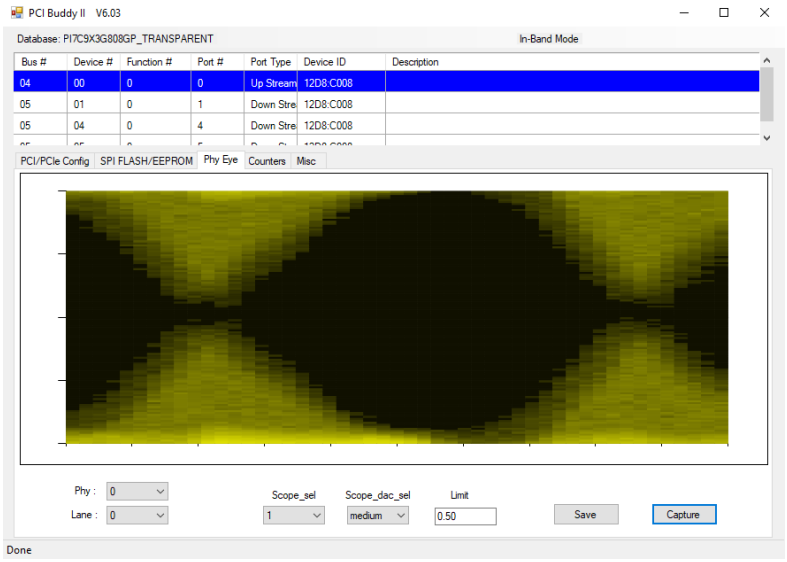


3) 380h = 33333333 (check by 3C8h)
1N11 = 7404h (Enable ADAPT_AFE & ADAPT_DFE)

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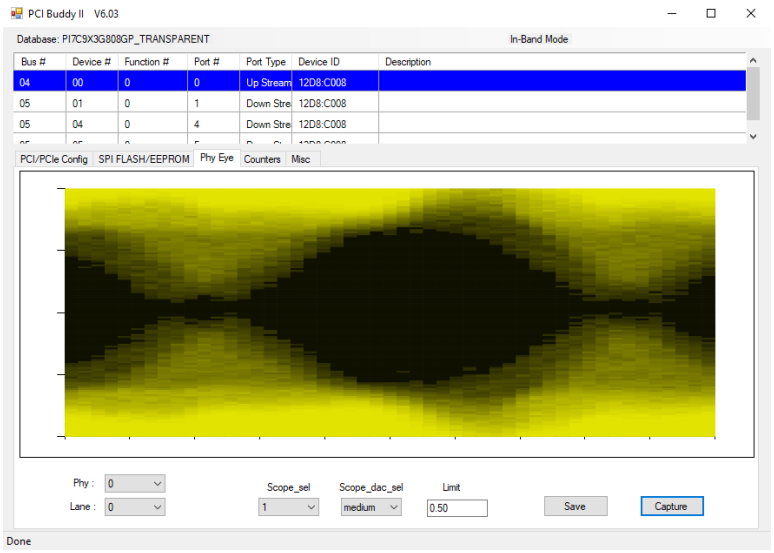


1N11 = 4404 (Disable ADAPT_AFE & ADAPT_DFE)

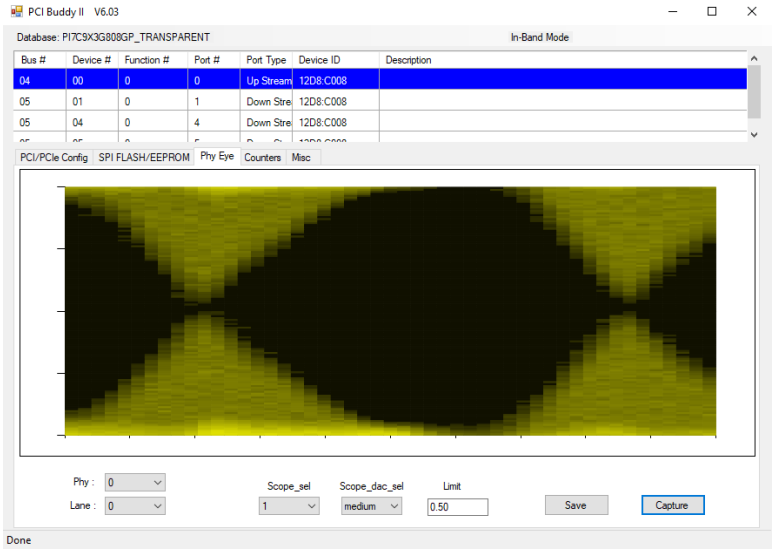


4) 380h = 44444444 (check by 3C8h)
1N11 = 7404h (Enable ADAPT_AFE & ADAPT_DFE)

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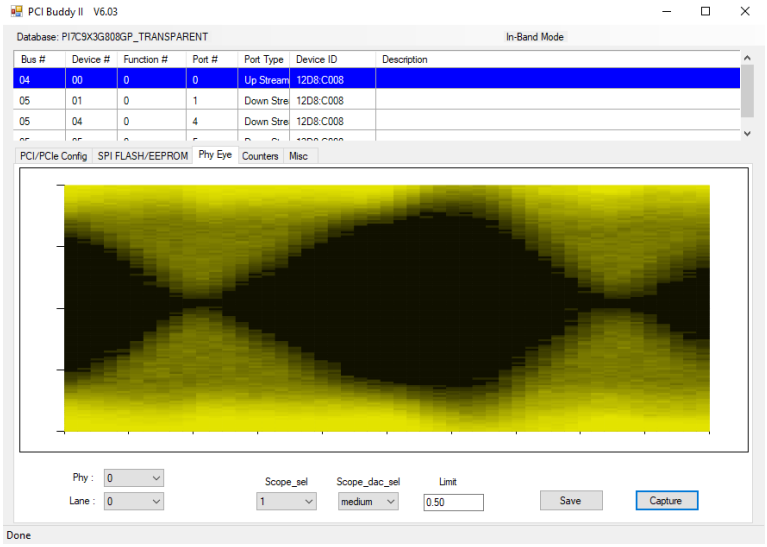


1N11 = 4404 (Disable ADAPT_AFE & ADAPT_DFE)

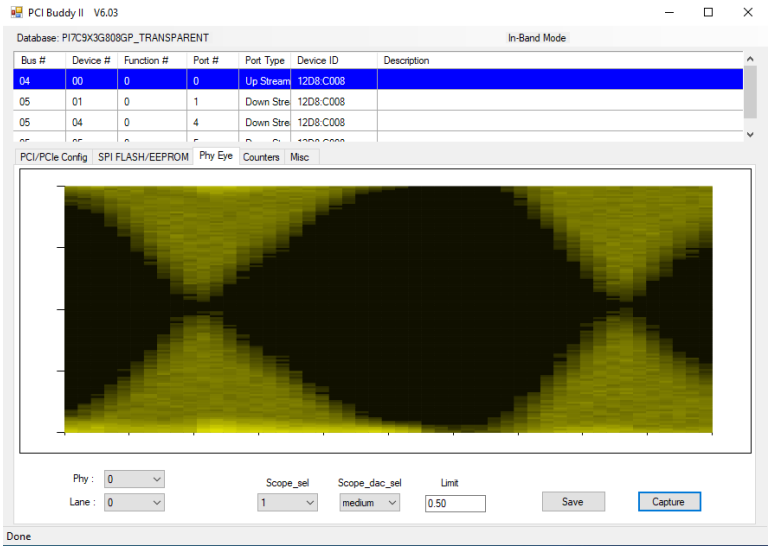


5) 380h = 55555555 (check by 3C8h)
1N11 = 7404h (Enable ADAPT_AFE & ADAPT_DFE)

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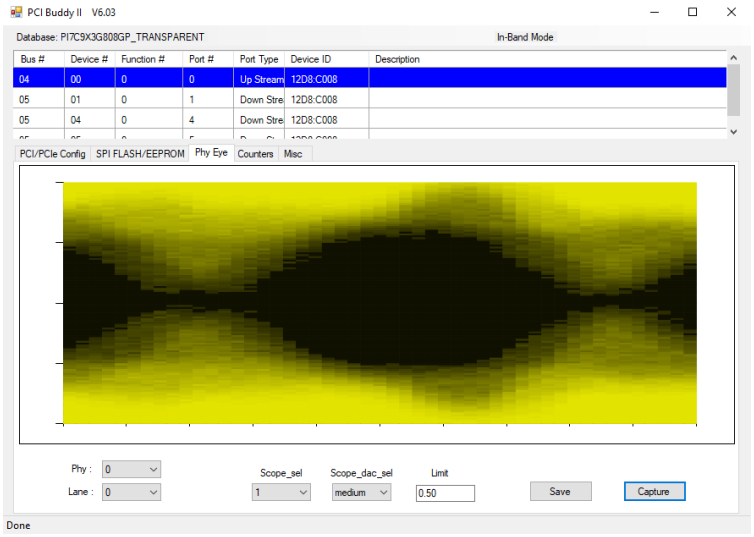


1N11 = 4404 (Disable ADAPT_AFE & ADAPT_DFE)

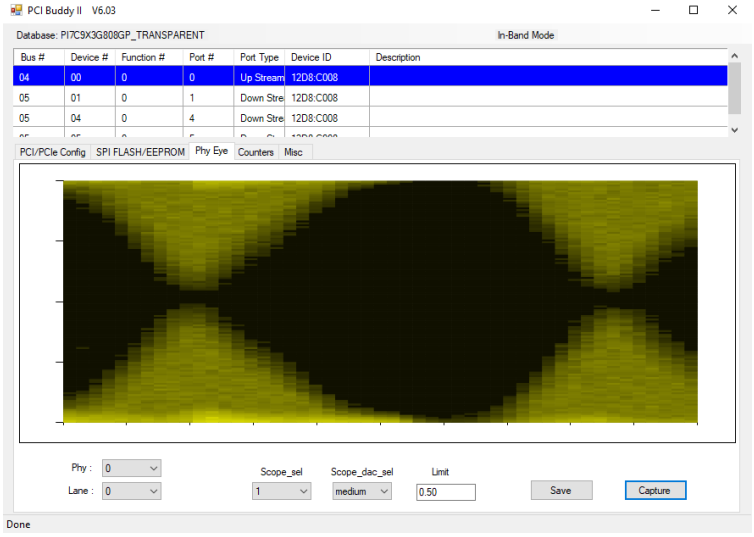


6) 380h = 66666666 (check by 3C8h)
1N11 = 7404h (Enable ADAPT_AFE & ADAPT_DFE)

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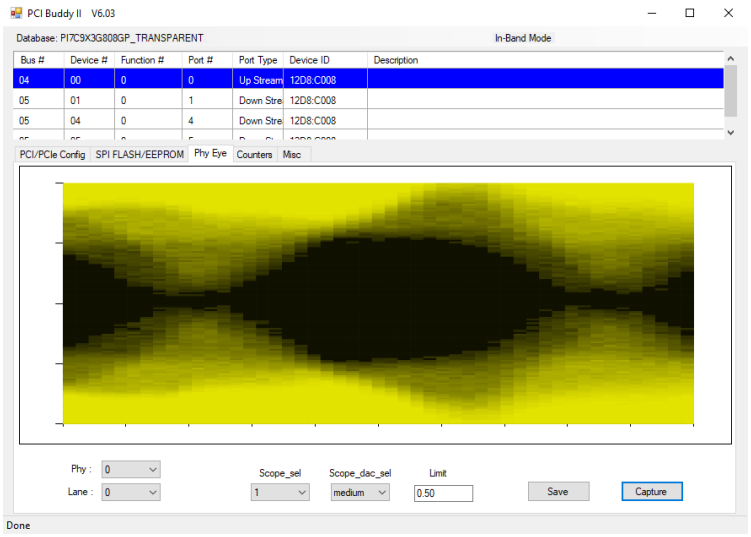


1N11 = 4404 (Disable ADAPT_AFE & ADAPT_DFE)

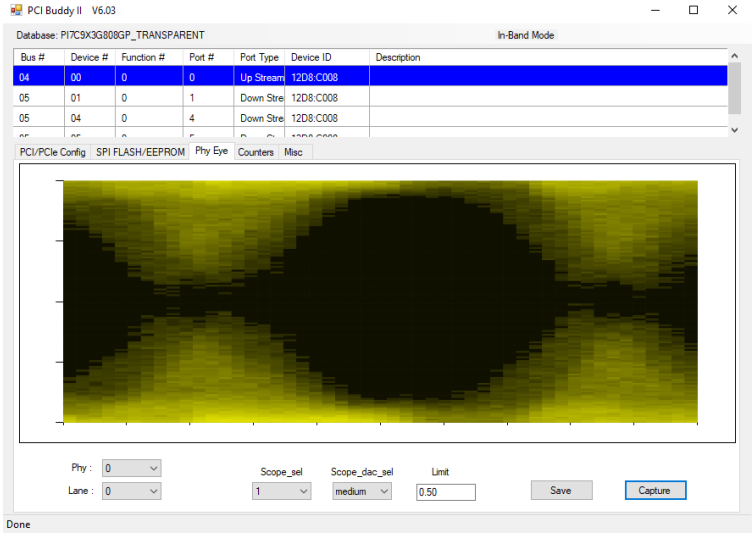


7) 380h = 77777777 (check by 3C8h)
1N11 = 7404h (Enable ADAPT_AFE & ADAPT_DFE)

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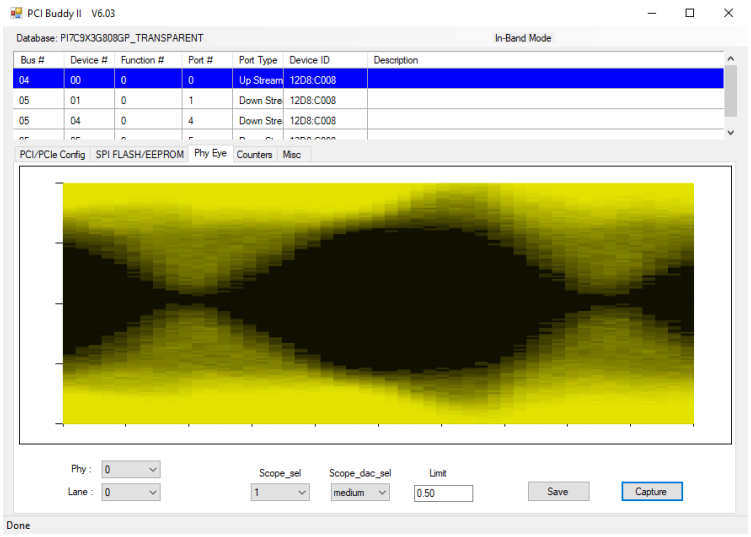


1N11 = 4404 (Disable ADAPT_AFE & ADAPT_DFE)

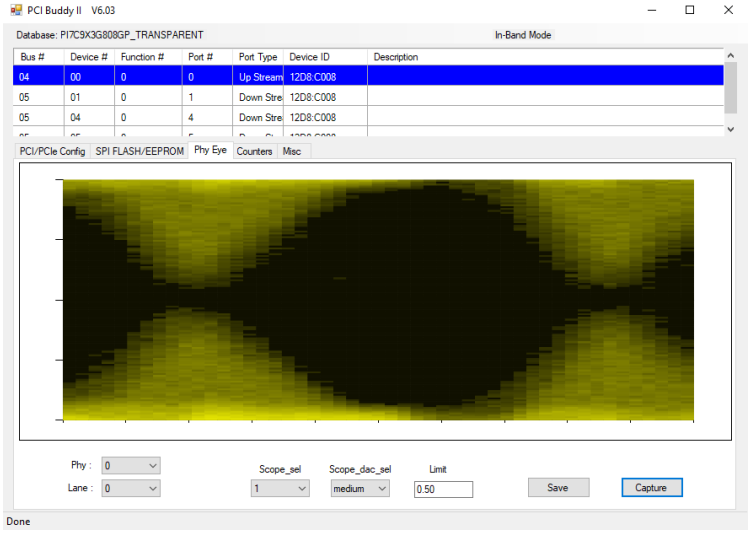


8) 380h = 88888888 (check by 3C8h)
1N11 = 7404h (Enable ADAPT_AFE & ADAPT_DFE)

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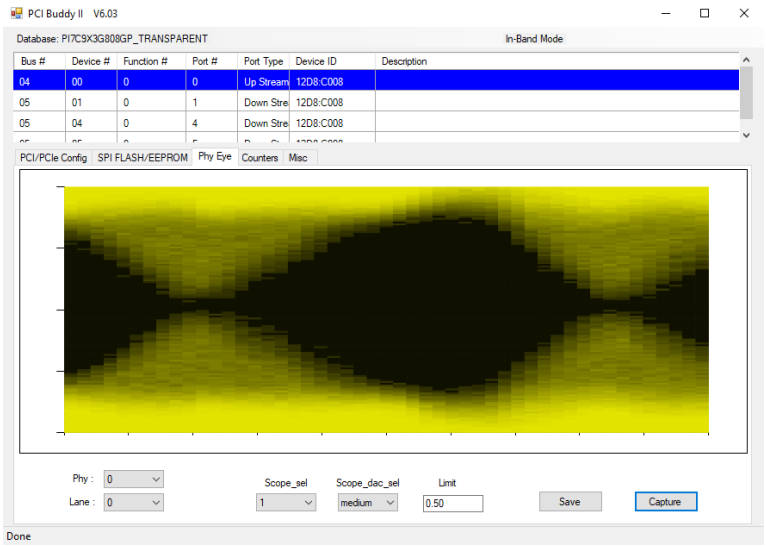


1N11 = 4404 (Disable ADAPT_AFE & ADAPT_DFE)

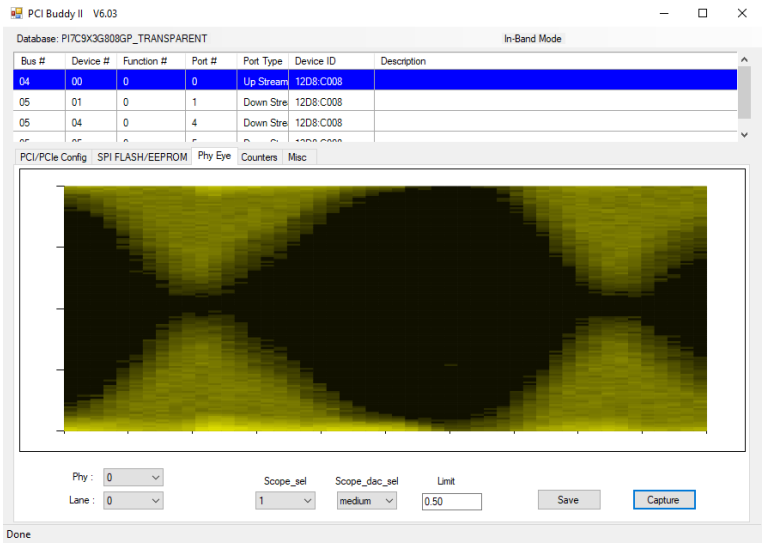


9) 380h = 99999999 (check by 3C8h)
1N11 = 7404h (Enable ADAPT_AFE & ADAPT_DFE)

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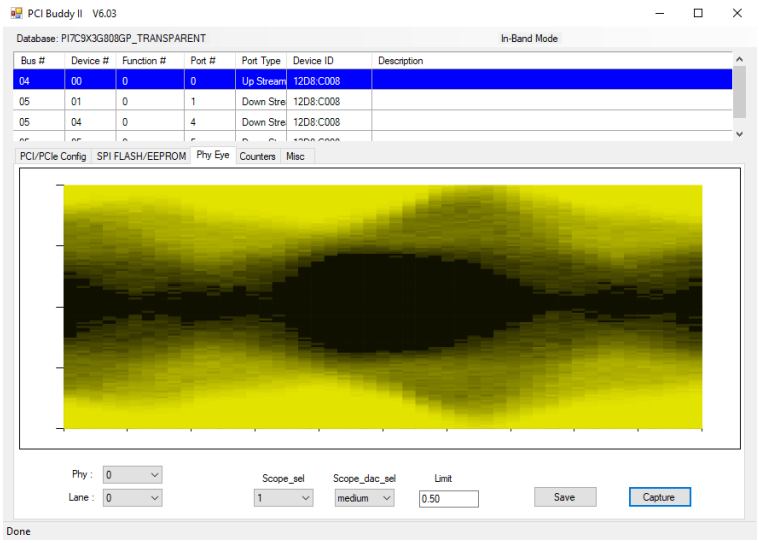


1N11 = 4404 (Disable ADAPT_AFE & ADAPT_DFE)

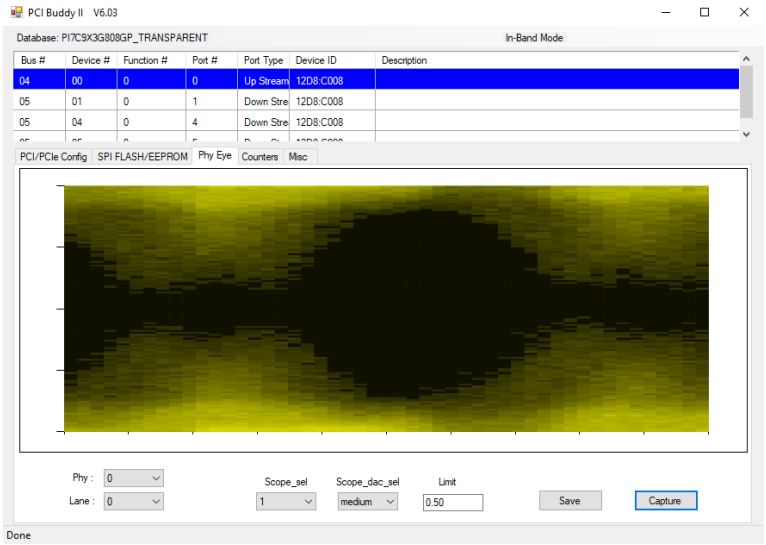


10) 380h = AAAAAAAA (check by 3C8h)
    1N11 = 7404h (Enable ADAPT_AFE & ADAPT_DFE)

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1N11 = 4404 (Disable ADAPT_AFE & ADAPT_DFE)



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15 INTA0_L and INTA1_L Trigger Sources

When INTA0_L/INTA1_L is driven low, it indicates that one or more of the following events/errors are detected: Hot Plug events, Link State events, General-Purpose Input Interrupt events, Device-Specific CDEP Port Link Interface errors and events, CDEP-Virtual Doorbell events or CDEP-Link Doorbell events. CFG offset 4C0h.bit[10:8] are used to select what is the trigger source for INTA0_L and INTA1_L. PI7C9X3G1632GP-3G1224GP includes two tiles, INTA0_L is used for Tile 0, and INTA1_L is used for Tile 1.

4C0h.bit[8]=1, INTA0_L/INTA1_L is triggered by slot interface, including hot plug events, and link status events

4C0h.bit[9]=1, INTA0_L/INTA1_L is triggered by GPIO

4C0h.bit[10]=1, INTA0_L/INTA1_L is triggered by CDEP interface, including Device-Specific CDEP Port Link Interface errors and events, CDEP-Virtual Doorbell events or CDEP-Link Doorbell events.

15.1 INTA0_L/INTA1_L is triggerred by Slot

When hot plug function is enabled, switch will generate INT or MSI message when hot plug or link state events are triggerred. If CFG offset 4Ch.bit[8] is set, switch will not generate INT or MSI message when hot plug or link state events are triggerred. Instead, it will drive INTA0_L/INTA1_L low and will not drive INTA0_L/INTA1_L low after status is clear (INTA0_L/INTA1_L will be high due to external pull-up resistor).

It is valided for down port only.

15.2 INTA0_L/INTA1_L is triggerred by GPIO

INTA0_L/INTA1_L can be triggerred by GPIO. Once GPIO is selected, the value of the corresponding bit according to the GPIO register need be changed.

Following flow describes what and how to program GPIO to trigger INTA0_L/INTA1_L signal. As only P0 can modify GPIO registers, all commands are issued for P0 only.

Following “x” in each step means the same GPIO pin. For example, if select GPIO[0] as the trigger source, x is “0”.

1) Set INTA0_L/INTA1_L is triggered by GPIO
   Port 0, offset 4C0h.bit[9]=1b
2) Set GPIO input pin is de-bounced
   Port 0, offset 324h bit[x]=1b
3) Set GPIO input pin is active-high
   Port 0, offset 338h bit[x]=1b
4) Set GPIO interrupt is unmasked
   Port 0, offset 340h.bit[x]=0b
5) Measure INTA0_L/INTA1_L signal level when set GPIO input as 1 or 0
   a) Set Port 0, offset 31Ch.bit[2x-1]=1b, INTA0_L/INTA1_L should be high
   b) Set Port 0, offset 31Ch.bit[2x-1]=0b, INTA0_L/INTA1_L should be low

15.3 INTA0_L/INTA1_L is triggerred by CDEP Interface

When CDEP function is enabled, INTA0_L/INTA1_L will be triggerred by Device-Specific CDEP Port Link Interface errors and events, CDEP-Virtual Doorbell events or CDEP-Link Doorbell events.

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16 Lane Reversal Support

This switch upport Lane Reversal, but have some limitation. Please refer to below table for support list.

Table 16-1 Lane Reversal Support Table

Packet Switch[*1] Lane Width	Link Partner[*2] Lane Width	Lane Number Mapping	
		Packet Switch	Link Partner
x16	x16	Lane 0 ~ Lane 15	Lane 15 ~ Lane 0
x16	x8	Lane 0 ~ Lane 7	Lane 7 ~ Lane 0
x16	x8	Lane 8 ~ Lane 15	Lane 7 ~ Lane 0
x16	x4	Lane 0 ~ Lane 3	Lane 3 ~ Lane 0
x16	x4	Lane 12 ~ Lane 15	Lane 3 ~ Lane 0
x16	x2	Lane 0 ~ Lane 1	Lane 1 ~ Lane 0
x16	x2	Lane 14 ~ Lane 15	Lane 1 ~ Lane 0
x16	x1	Lane 15	Lane 0
x8	x8	Lane 0 ~ Lane 7	Lane 7 ~ Lane 0
x8	x4	Lane 0 ~ Lane 3	Lane 3 ~ Lane 0
x8	x4	Lane 4 ~ Lane 7	Lane 3 ~ Lane 0
x8	x2	Lane 0 ~ Lane 1	Lane 1 ~ Lane 0
x8	x2	Lane 6 ~ Lane 7	Lane 1 ~ Lane 0
x8	x1	Lane 7	Lane 0
x4	x4	Lane 0 ~ Lane 3	Lane 3 ~ Lane 0
x4	x2	Lane 0 ~ Lane 1	Lane 1 ~ Lane 0
x4	x2	Lane 2 ~ Lane 3	Lane 1 ~ Lane 0
x4	x1	Lane 3	Lane 0
x2	x2	Lane 0 ~ Lane 1	Lane 1 ~ Lane 0
x2	x1	Lane 1	Lane 0

Notes:
1. Packet Switch means Diodes GEN3 Serial Product. Lane Reversal can be either upstream port or downstream port of Diodes Packet Switch.
2. Link Partner may be Root or End device.
3. Lane reversal TX/RX must be flipped at the same time, not just flip TX or RX.
4. Lane reversal and Polarity Inversion can co-exist.
5. Connection methods not listed in the table are not supported.

By default setting (offset 3ACh.bit[20:19]=01b), if switch's lane detection result is the same as the lane number it receives, it will perform "lane reversal" function; if they are different, switch will not do lane-reversal. Sometimes, RC/EP may not link up at lane-reversal up/down port if the lane detection result is different to the received lane number. Please try to set offset 3ACh.bit[20:19]=00b, then switch will perform "lane reversal" function based on "receive lane number".

Offset 3ACh.bit[20:19] can be changed by I2C/SMBUS or EEPROM.

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17 Online Compliance Test Method

This switch support online compliance testing when it is difficult to use standard test fixtures (CBB / CLB).

Table 17-1 Online Compliance Test Method for Upstream Port

	Method (Link Control Register[4] and Link Control Register2[4:0])		
	Test Step	**description**	**Test Result**
RC + UP	Read UP 98h = 0043	default target link speed is GEN3	
	write RC 98h[4] = 1	set enter compliance	
	write RC 78h[4] = 1	link disable	
	write RC 78h[4] = 0	link enable, into compliance mode	
	write UP 98h[4:0] = 51	change gen1 speed and enter compliance	˅(GEN1 Compliance)
	write UP 98h[4:0] = 52 , [6]= 0	change gen2 speed and enter compliance	˅(GEN2 Compliance) -3.5db de-emphasis
	write UP 98h[4:0] = 52 , [6]= 1	change gen2 speed and enter compliance	˅(GEN2 Compliance) -6.0db de-emphasis
	write UP 98h[4:0] = 53	change gen2 speed and enter compliance	˅(GEN3 Compliance)*

Note.* Only support one type pattern. It does not change "Compliance Pattern Type" in Gen3

Table 17-2 Online Compliance Test Method for Downstream Port

	Method (Link Control Register[4] and Link Control Register2[4:0])		
	Test Step	**description**	**Test Result**
Port Empty	Read P7 98h = 0043	default target link speed is GEN3	
	write P7 98h[4] = 1	set enter compliance	
	write P7 78h[4] = 1	link disable	
	write P7 78h[4] = 0	link enable, into compliance mode	X (no compliance)
	write P7 98h[4:0] = 51	set gen1 speed and enter compliance	
	write P7 78h[4] = 1	link disable	X (now is in polling state, can be not enabled!)
Intel Pro 1000 (GEN1)	Read P7 98h = 0043	default target link speed is GEN3	
	write P7 98h[4] = 1	set enter compliance	
	write P7 78h[4] = 1	link disable	
	write P7 78h[4] = 0	link enable, into compliance mode	˅(GEN3 Compliance)*
	write P7 98h[4:0] = 51	change gen1 speed and enter compliance	˅(GEN1 Compliance)
Marvell SATA 9215 (GEN2)	Read P7 98h = 0043	default target link speed is GEN3	
	write P7 98h[4] = 1	set enter compliance	
	write P7 78h[4] = 1	link disable	
	write P7 78h[4] = 0	link enable, into compliance mode	˅(GEN3 Compliance)*
	write P7 98h[4:0] = 51	change gen1 speed and enter compliance	˅(GEN1 Compliance)
	write P7 98h[4:0] = 52 , [6]= 0	change gen2 speed and enter compliance	˅(GEN2 Compliance) -3.5db de-emphasis
	write P7 98h[4:0] = 52 , [6]= 1	change gen2 speed and enter compliance	˅(GEN2 Compliance) -6.0db de-emphasis
Intel 760 SSD (GEN3)	Read P7 98h = 0043	default target link speed is GEN3	
	write P7 98h[4] = 1	set enter compliance	
	write P7 78h[4] = 1	link disable	
	write P7 78h[4] = 0	link enable, into compliance mode	˅(GEN3 Compliance)*
	write P7 98h[4:0] = 51	change gen1 speed and enter compliance	˅(GEN1 Compliance)
	write P7 98h[4:0] = 52 , [6]= 0	change gen2 speed and enter compliance	˅(GEN2 Compliance) -3.5db de-emphasis
	write P7 98h[4:0] = 52 , [6]= 1	change gen2 speed and enter compliance	˅(GEN2 Compliance) -6.0db de-emphasis

Note.* Only support one type pattern. It does not change "Compliance Pattern Type" in Gen3

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Revision History

Date	Revision	Description
2020/7/24	0.1	Preliminary Applicatioin Note
2020/7/29	0.2	Update GPIO Chaper
2020/8/12	0.3	Add Table of Content Add List of Table Update GPIO Chapter Delete Hot Plug Chapter Update Thermal Sensor Access Guidelines Chapter Add Master Loopback Chapter
2020/8/31	0.4	Add List of Figure Update GPIO Chapter Add IREF Chapter Fix Typo
2020/12/23	0.5	Update chapter 4 and change title as CLOCK Scheme Add SHP Chapter
2021/03/17	0.6	Add 3.10 Total loss budgets Add chapter 4 Supplies Requirements Add chapter 12 JTAG Pins
2021/07/16	0.7	With “0.5 modify by JC” as the main, combined with the data of “0.6_eros_0629” Re-arrange the chapters Add Chapter Reduction of Crosstalk in High Speed Material PCB Add Chapter PHY TX/RX Signal Adjustment Add Chapter Troubleshooting
2021/11/24	0.8	Re-arrange the chapters Add Chapter 15 INTA0_L/INTA1_L Trigger Source Add Chapter 16 Lane Reversal Support Remove Chapter “Troubleshooting” to Chapter 17 Update some typo
2023/07/06	0.9	Re-arrange the chapters Add Chapter 18 Online Compliance Test Method Add chapter 14-3 To disable Adapt_DFE_EN and Adapt_AFE_EN to check RX eye at different preset Update chapter 7.2 and 7.3 Remove Troubleshooting section
2024/08/20	0.a	Update Table 6-1 Update Table 7-1 Update Table 7-2 Update Table 7-3 Update Table 7-4 Update Table 9-1 Update Table 11-1 Update Table 12-3 Update Table 13-1
2024/10/29	0.b	Add PI7C9X3G1224GP Table 5 2 Power Supply Requirements for PI7C9X3G1224GP Change the file name to support all GEN3 parts

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